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Inventor(s)

Schmidt; Fabio et al.

Hearing device comprising a mechanical element for sound transmission

Abstract

The disclosure relates to a hearing device configured to be worn at an ear of a user, the hearing device including a sound delivery (SD) part configured to be at least partially inserted into an ear canal and having a custom shell at least partially customized to the ear canal and an electroacoustic transducer housed within the custom shell. The hearing device further includes a mechanical element connected to a medial end of the custom shell of the SD part, the mechanical element having a structure enclosing a cavity with an opening facing the SD part such that the cavity is acoustically coupled to the electroacoustic transducer. The structure includes a section in which thinner portions and thicker portions are alternately arranged around an axis of the structure.

Inventors: Schmidt; Fabio (Uetikon am See, CH), Römmeler; Arno (Zürich, CH), Kuipers; Erwin (Stäfa, CH), Vonlanthen; Andi (Oberrohrdorf, CH)

Applicant: SONOVA AG (Stäfa, CH)

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Background/Summary

RELATED APPLICATIONS [0001] The present application is a continuation-in-part of U.S. patent application Ser. No. 18/932,386, filed Oct. 30, 2024, which claims priority to EP Patent Application No. 23218773.2, filed Dec. 20, 2023, each of which is hereby incorporated by reference in its entirety.

BACKGROUND INFORMATION

[0002] Hearing devices may be used to improve the hearing capability or communication capability of a user, for instance by compensating a hearing loss of a hearing-impaired user, in which case the hearing device is commonly referred to as a hearing instrument such as a hearing aid, or hearing prosthesis. A hearing device may also be used to output sound based on an audio signal which may be communicated by a wire or wirelessly to the hearing device. A hearing device may also be used to reproduce a sound in a user's ear canal detected by an input transducer such as a microphone or a microphone array. The reproduced sound may be amplified to account for a hearing loss, such as in a hearing instrument, or may be output without accounting for a hearing loss, for instance to provide for a faithful reproduction of detected ambient sound and/or to add audio features of an augmented reality in the reproduced ambient sound, such as in a hearable. A hearing device may also provide for a situational enhancement of an acoustic scene, e.g. beamforming and/or active noise cancelling (ANC), with or without amplification of the reproduced sound. A hearing device may also be implemented as a hearing protection device, such as an earplug, configured to protect the user's hearing. Different types of hearing devices configured to be worn at an ear include earbuds, earphones, hearables, and hearing instruments such as receiver-in-the-canal (RIC) hearing aids, behind-the-ear (BTE) hearing aids, in-the-ear (ITE) hearing aids, invisible-in-the-canal (IIC) hearing aids, completely-in-the-canal (CIC) hearing aids, cochlear implant systems configured to provide electrical stimulation representative of audio content to a user, a bimodal hearing system configured to provide both amplification and electrical stimulation representative of audio content to a user, or any other suitable hearing prostheses. A hearing system comprising two hearing devices configured to be worn at different ears of the user is sometimes also referred to as a binaural hearing device. A hearing system may also comprise a hearing device, e.g., a single monaural hearing device or a binaural hearing device, and a user device, e.g., a smartphone and/or a smartwatch, communicatively coupled to the hearing device.

[0003] Typically, the gain of the sound amplified by a hearing device depends on frequency. Such a dependency is commonly referred to as the frequency response of the hearing device. The frequency response is a crucial parameter for listening experience, ensuring a high quality in sound reproduction, e.g., with regard to a naturalness, clarity, and consistency of the reproduced sound, and, in a hearing instrument, for allowing a good fitting of the sound amplification to an individual hearing loss. In particular, a broad bandwidth of the frequency response is often desirable. Further parameters affecting the audio quality include the number and position of resonances in the frequency response spectrum and the flatness of the frequency response curve. For instance, on the one hand, a rather flat frequency response can ensure that all frequencies are reproduced equally. On the other hand, a boosting or attenuating of certain frequencies or frequency ranges may be

useful to implement targeted adjustments in the frequency output, e.g., to counteract imbalances or distortions.

[0004] The frequency response of a hearing device can be affected by many parameters including the properties of various components included in the hearing device, e.g., an amplifier, receiver, sound tube, filter, vent, ear canal sealing and/or the like, and the way these components are arranged and interact. Nevertheless, the choices in the design of a hearing device are also limited by size constrictions in order to be wearable at the ear, in particular to fit into an ear canal of a person. A further impact on the frequency response has the ear canal geometry itself which individually varies, at least to a certain extent. These constraints in the design of the hearing device may limit a desired frequency response behavior. It would therefore be desirable to provide for a means allowing to shape or tweak the frequency response governed by those constraints toward the more desired behavior.

[0005] Another problem of hearing devices which are at least partially worn inside an ear canal is that substances such as cerumen, liquids or dirt may enter a transducer, e.g. a receiver, or a sound tube connected to the transducer. Such an ingress may result in transducer malfunctioning and deterioration in hearing performance ranging from slightly distorted acoustic signals to a total failure of the transducer. Current solutions include so-called wax filters or cerumen filters preventing the substances from entering either by a more or less dense grid, or a membrane. One type of those transducer protection elements may be characterized as an acoustically open system allowing a transition of sound with rather small impact and/or acceptable changes of the sound output. Typical implementations comprise a fine and dense mesh that blocks cerumen, for instance at the medial side in front of a receiver. However, cerumen has a certain ability to stick to such a filter and cause partial or complete clogging of the filter. In case the level of clogging is too high, the hearing device wearer will perceive reduced and possibly distorted acoustic signals. This may require exchanging of the filter during which, however, the cerumen may be pushed even further inside.

[0006] Another type of transducer protectors may be characterized as acoustically closed which may be implemented, e.g., as a small flat membrane placed in front of the transducer. The use of such a small flat membrane has the disadvantage that, in particular at large sound levels, the membrane displacement is a nonlinear function of the acoustic pressure leading to nonlinearities in the reproduced sound in the ear canal. Another disadvantage is, because the membrane needs to be very compliant, that the membrane can easily get damaged or punctured. An alleviation of this issue would require an increase of the membrane's bending stiffness, e.g., by a thickening or changing the material properties, or of the membrane's pre-tension. Both aspects would lead to a substantial sound transmission loss. Such loss may be compensated by larger receivers which would, however, increase the size of the hearing device and thus result in a lower fit-rate of the hearing device into an ear canal.

[0007] US 2021/0258705 A1 discloses a transducer protection element including a dome implemented as a curved sound radiating part, a neck part for mounting the element to the transducer, and a suspension part formed between the dome and the neck part along a movement direction of the dome so as to provide for sound passage through the dome with little attenuation and distortion in a way that the protection element is acoustically transparent with regard to its impact on the transmitted sound. However, also with regard to unfavorable or improvable sound properties, the amplified sound can thus not be altered by its transmission through the protection element.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings. The drawings illustrate various embodiments and are a part of the specification. The illustrated embodiments are merely examples and do not limit the scope of the disclosure. Throughout the drawings, identical or similar reference numbers designate identical or similar elements. In the drawings:

[0009] FIG. 1 schematically illustrates an exemplary hearing device including a SD part connected to a mechanical element;

[0010] FIG. 2 schematically illustrates another exemplary hearing device including a SD part connected to a mechanical element;

[0011] FIGS. 3A, 3B schematically illustrate arrangements in which a hearing device including a SD part is at least partially inserted into an ear canal, wherein a mechanical element is connected or disconnected;

[0012] FIGS. 4A-4D schematically illustrate another exemplary hearing devices including a SD part connected to a mechanical element

[0013] FIGS. 5A-5H schematically illustrate circumferences of a structure of a mechanical element;

[0014] FIGS. 6A-6D schematically illustrate mechanical elements;

[0015] FIGS. 7A, 7B schematically illustrate further mechanical elements;

[0016] FIGS. 8A-8D schematically illustrate another mechanical element;

[0017] FIGS. 9A-9D schematically illustrate another mechanical element;

[0018] FIGS. 10A, 10B schematically illustrate another mechanical element;

[0019] FIGS. 11A, 11B schematically illustrate another mechanical element;

[0020] FIGS. 12A-12C schematically illustrate further mechanical elements;

[0021] FIGS. 13A, 13B schematically illustrate a thinner portion of a structure of a mechanical element during sound transmission;

[0022] FIG. 14 illustrates an exemplary frequency response curve;

[0023] FIGS. 15A-15B schematically illustrate an exemplary hearing device including a SD part integral with a mechanical element;

[0024] FIGS. 16A-16B schematically illustrate another exemplary hearing device including a SD part integral with a mechanical element;

[0025] FIGS. 17A-17B schematically illustrate an exemplary mold for forming hearing devices;

[0026] FIG. 18 schematically illustrates another exemplary mold for forming hearing devices; and

[0027] FIG. 19 illustrates an exemplary method for forming hearing devices using a mold.

DETAILED DESCRIPTION

[0028] Described herein are a hearing device comprising a sound delivery (SD) part configured to be at least partially inserted into an ear canal, and a mechanical element configured to be connected to the SD part, according to the preamble of claim 1.

[0029] It is a feature of the present disclosure to avoid at least one of the above mentioned disadvantages and to provide for a hearing device in which properties of the sound outputted by an SD part can be modified in a convenient and/or adaptable and/or cost-efficient and/or user-friendly manner, e.g., with regard to a frequency response and/or a pressure of the outputted sound. It is another feature to provide for a hearing device with reliable protection against an ingress of substances. It is a further feature to provide for a mechanical element included in the hearing device which can be designed for the altering of sound outputted by the sound emitter in a desired way, e.g., with regard to possible improvements of the outputted sound, and which is also adapted for ingress protection. It is yet another feature to provide for a hearing device with an improved acoustic performance and/or good ingress protection and/or favorable geometrical dimensions, in particular with regard to a smaller size of the hearing device and/or a reduced length of a sound

tube included in the hearing device.

[0030] Accordingly, the present disclosure proposes a hearing device configured to be worn at an ear of a user, the hearing device comprising [0031] a sound delivery (SD) part configured to be at least partially inserted into an ear canal, the SD part comprising a sound tube having an axis in which direction sound is propagating, wherein the sound tube is acoustically coupled to an electroacoustic transducer; and [0032] a mechanical element configured to be connected to the SD part, the element comprising a structure enclosing a cavity with an opening facing the SD part when the element is connected to the SD part such that the cavity is acoustically coupled to the electroacoustic transducer, wherein the structure comprises a section in which thinner portions and thicker portions are alternatingly arranged around an axis of the structure, wherein the axis of the structure is aligned within 20° relative to the axis of the sound tube when the element is connected to the SD part.

[0033] Thus, by acoustically connecting the mechanical element to the SD part, the thinner portions provided at a circumference surrounding the axis of the structure can be employed to provide for a transmission of the emitted sound in an advantageous way. The thinner portions may more easily be stimulated to vibrate during sound transmission, e.g., with a smaller mechanical damping and/or a larger vibration amplitude, as compared to the thicker portions due to a smaller mass and/or stiffness of the thinner portions. Compared to flat membranes known in the art, a larger surface can be realized by the structure of the mechanical element, resulting in improved sound transmission. Further, a volume of the cavity enclosed by the structure can be selected to be small enough to influence the acoustical properties in a desired way. In addition, the thinner and thicker portions can be geometrically designed to modify a frequency response and/or a pressure of the outputted sound in a desired way. A required acoustic transmission performance may be achieved by the combination of several thinner portions, whereas the mechanical stability is provided by different, thicker portions. Moreover, the mechanical element can serve as a protector against an ingress of substances into apertures at the SD part, e.g., at the sound tube. In this way, a functionality of an earwax filter may be implemented. Replacing an earwax filter with the mechanical element can be exploited for a further advantage to reduce a required length of the SD part, e.g., a sound tube, in which such an earwax filter would be commonly integrated.

[0034] Subsequently, additional features of the hearing device are described. Each of those features can be provided solely or in combination with at least another feature.

[0035] In some implementations, the structure is configured so as to provide for, when the element is connected to the SD part, a frequency response of the hearing device differing from the frequency response when the element is disconnected from the SD part.

[0036] In some implementations, the structure is configured to provide for, when the element is connected to the SD part, an acoustic pressure inside the ear canal which is increased for at least one frequency comprised in a frequency range between 1 kHz and 10 kHz as compared to the acoustic pressure when the element is disconnected. In some implementations, the acoustic pressure is increased by at least 2 dB, e.g., at least 3 dB, or at least 4 dB, or at least 5 dB. In some implementations, the acoustic pressure is increased within a frequency bandwidth comprised in said frequency range, wherein the frequency bandwidth is at least 10 Hz, e.g., 50 Hz.

[0037] In some implementations, the structure is configured to provide for, when the element is connected to the SD part, an acoustic pressure inside the cavity exceeding an acoustic pressure outside the cavity inside ear canal. In some implementations, the acoustic pressure inside the cavity exceeds the acoustic pressure outside the cavity by at least 5 dB in a frequency range between 100 Hz and 10 kHz.

[0038] In some implementations, the structure is configured to provide for at least one resonance frequency which, when the element is connected to the SD part, is shifted in the frequency response of the hearing device as compared to when the element is disconnected. In some implementations, the resonance frequency is shifted by at least 50 Hz.

[0039] In some implementations, the resonance frequency, when the element is connected to the SD part, is between 2 kHz and 7 kHz.

[0040] In some implementations, the frequency response comprises two resonance frequencies in a frequency range, e.g., between 1 kHz and 10 kHz, in particular between 2 kHz and 9 kHz, when the element is disconnected, and three resonance frequencies in this frequency range when the element is connected.

[0041] In some implementations, the structure is configured to provide for a cutoff frequency of at least 8 kHz when the element is connected. The cutoff frequency may be defined as a frequency above which a sound pressure level (SPL), in particular an SPL measurable at the tympanic membrane, is reduced by 20 dB per octave.

[0042] In some implementations, the electroacoustic transducer comprises a balanced armature transducer and/or a moving coil driver.

[0043] In some implementations, the thinner portions and the thicker portions are alternately arranged around the axis of the structure such that a transmission of sound through the structure is more effective at the thinner portions and a rigidity of the structure is more enhanced by the thicker portions.

[0044] In some implementations, said section of the structure is provided such that a volume of the enclosed cavity is minimized and a circumference along which the thinner portions and thicker portions are alternately arranged, e.g., an outer surface of the circumference, is maximized.

[0045] In some implementations, the thinner portions constitute at least two third of a circumference of the structure around which the thinner portions and the thicker portions are alternately arranged.

[0046] In some implementations, the circumference of the structure comprises at least six thinner portions. In some implementations, the circumference of the structure comprises at least seven thinner portions. In some implementations, the circumference of the structure comprises at least eight thinner portions.

[0047] In some implementations, a natural frequency of at least two of the thinner portions differs. The natural frequency may also be denoted as the eigenfrequency. In some implementations, the natural frequencies differ by at least 10 Hz, e.g., at least 50 Hz. Each of the different natural frequencies may thus contribute to a resonance frequency of the hearing device when the mechanical element is connected to the SD part. E.g., an occurrence of a sharp resonance peak may thus be avoided. In some implementations, the different natural frequencies are selected such that they collectively provide for a single peak in a frequency response spectrum when the mechanical element is connected to the SD part.

[0048] In some implementations, each of the thicker portions is arranged between two of the thinner portions. In some implementations, each of the thicker portions is protruding at an inner and/or outer surface of the structure.

[0049] In some implementations, one or more of the thinner portions arranged around the axis of the structure, e.g., each of the thinner portions arranged around the axis of the structure, are provided as a portion of the structure not exceeding, i.e. falling below or being equal, a thickness limit. In some implementations, each of the thicker portions is a portion exceeding, i.e. not falling below and not being equal, the thickness limit. In some implementations, the thickness limit is selected as a value of 0.15 mm or smaller. In some implementations, the thickness limit is a value of 0.14 mm or smaller, e.g., 0.13 mm or smaller, 0.12 mm or smaller, 0.11 mm or smaller, 0.10 mm or smaller, 0.09 mm or smaller, 0.08 mm or smaller, 0.07 mm or smaller, 0.06 mm or smaller, 0.05 mm or smaller, 0.04 mm or smaller, 0.03 mm or smaller, 0.02 mm or smaller, or 0.01 mm or smaller. While a smaller value of the thickness limit may be desirable in some implementations, providing such a thickness limit, e.g., below 0.1 mm and/or below 0.05 mm, may be restricted, e.g., due to production constraints.

[0050] In some implementations, one or more of the thicker portions arranged around the axis of

the structure, e.g., each of the thicker portions arranged around the axis of the structure, has a maximum thickness which is at least 0.03 mm larger than a maximum thickness of a thinner portion adjoining, e.g., bordering, the thicker portion. In some implementations, one or more of the thicker portions arranged around the axis of the structure has a maximum thickness which is at least 0.05 mm larger than a maximum thickness of a thinner portion adjoining the thicker portion. In particular, selecting the maximum thickness of the thicker portions relative to the maximum thickness of an adjoining thinner portion can allow to provide for a desired stability of the element.

[0051] To illustrate, in a specific example, one or one or more of the thicker portions arranged around the axis may have a maximum thickness of about 0.2 mm and one or one or more of the thinner portions arranged around the axis may have a maximum thickness of about 0.14 mm. In another example, one or one or more of the thicker portions arranged around the axis may have a maximum thickness of about 0.1 mm and one or one or more of the thinner portions arranged around the axis may have a maximum thickness of about 0.05 mm.

[0052] In some implementations, at least one of the thinner portions has a homogenous thickness. In some implementations, at least one of the thinner portions has a varying thickness. In some implementations, at least one of the thicker portions has a homogenous thickness. In some implementations, at least one of the thicker portions has a homogenous thickness.

[0053] In some implementations, a thickness of the structure progressively (e.g., continuously and/or gradually) decreases in one or more of the thicker portions toward one or more of the thinner portions.

[0054] In some implementations, at least one of the thinner portions has a flat outer and/or inner surface. In some implementations, at least one of the thinner portions has a curved outer and/or inner surface. In some implementations, the curvature of the at least one thinner portion is smaller than the curvature of a tangent curve, e.g., an elliptical or circular tangent curve, touching a cross section of the structure.

[0055] In some implementations, the structure has a cross section with edges joined at corners, wherein the corners are formed by at least part of the thicker portions and the edges comprise at least part of the thinner portions.

[0056] In some implementations, at least two of the thinner portions are angled relative to one another.

[0057] In some implementations, the thinner portions have a width between adjoining thicker portions, and the thicker portions have a width between adjoining thinner portions, wherein the width of at least one of the thinner portions exceeds the width of the adjoining thicker portions by at least a factor of two. In some implementations, the width of at least one of the thinner portions exceeds the width of the adjoining thicker portions by at least a factor of three.

[0058] In some implementations, the structure encloses a volume inside the ear canal between the SD part and the tympanic membrane.

[0059] In some implementations, the structure has a cross section approximating a polygon. In some implementations, the polygon is concave. In some implementations, the polygon is convex. In some implementations, the polygon has a star shape.

[0060] In some implementations, at least two of the thinner portions, in particular thinner portions neighbouring each other, span an angle of at most 120° at the circumference of the structure. In some implementations, at least two of the thinner portions span an angle of at most 90° at the circumference of the structure. In some implementations, at least two of the thinner portions span an angle of at most 60° at the circumference of the structure. In some implementations, at least two of the thinner portions span an angle of at most 50° at the circumference of the structure.

[0061] In some implementations, at least two of the thinner portions lead to a corner at the circumference of the structure, wherein at least one of the thicker portions is arranged at the corner.

[0062] In some implementations, the structure has a cross section with a plurality of projections extending in a radial direction, wherein each of the projections comprises at least two thinner

portions. In some implementations, the structure has a cross section with a plurality of projections extending in a radial direction, wherein each of the projections comprises at least two thinner portions and a thicker portion arranged at a radially outer end of the projection. In some implementations, at least one of the projections has a shape of a jag and/or a lug and/or a ledge. [0063] In some implementations, the structure has a cross section approximating a shape of a star. In some implementations, the star shape is defined as a plurality of circularly arranged radial projections.

[0064] In some implementations, the structure has a cross section touching an outer tangent curve of a larger radius at which at least two of the thicker portions are located and an inner tangent curve of a smaller radius at which at least two other of the thicker portions are located, wherein at least two of the thinner portions extend between the outer tangent curve and the inner tangent curve. In some implementations, the outer and/or inner tangent curve is elliptical, in particular circular.

[0065] In some implementations, the structure has a cross section delimited inside an elliptical curve at which at least two of the thicker portions are located, wherein an inner volume enclosed by the structure is smaller than an outer volume enclosed between the circumference of the structure and the elliptical curve.

[0066] In some implementations, the mechanical element further comprises a sealing member configured for sealing against an ear canal wall, wherein the sealing member is integrally formed with the element. In some implementations, the sealing member has a shape of a dome.

[0067] In some implementations, the thinner portions constitute a surface of the structure of at least 30 mm.^{sup.2}. In some implementations, the thinner portions constitute a surface of the structure, e.g. a portion of the outer surface of the structure, of at least 40 mm.^{sup.2}, in particular at least 50 mm.^{sup.2}. In this way, a sound transmission through the protective element may be effectively increased, in particular by exploiting the 3D shape of the structure.

[0068] In some implementations, the structure comprises a front wall opposing the opening. In some implementations, the front wall comprises a thicker portion and/or a thinner portion. In some implementations, the structure comprises a lateral wall extending between the front wall and the opening, wherein a circumference surrounding the cavity is formed by the lateral wall. In some implementations, at least one of the thicker portions and/or at least one of the thinner portions of the lateral wall leads to the front wall.

[0069] In some implementations, one or more of the thicker portions protrude from the thinner portions between which they are arranged on the outside and/or on the inside of the structure. In some implementations, at least two of the thicker portions are spaced from one another along the circumference of the structure by one of the thicker portions arranged between and/or at least two of the thicker portions are spaced from one another along the circumference of the structure by one of the thinner portions arranged between.

[0070] In some implementations, one or more stiffeners are formed by one or more of the thicker portions. To illustrate, by providing the one or more thicker portions such that each one extends between two or more of the thinner portions of the structure, a stiffness of the structure can be increased between the thinner portions due to the larger rigidity of thicker portion forming the stiffener in between. In this way, an overall stability of the structure can be enhanced. In some implementations, one or more of the stiffeners are rod-shaped. In some implementations, one or more of the stiffeners are provided as fins. The fins may serve for reinforcement and/or for increasing the total sound radiating surface area and/or to obtain certain acoustic-mechanical vibration modes.

[0071] In some implementations, at least two of the thinner portions are spaced from one another along the circumference of the structure by a thicker portion arranged in between and/or at least two of the thicker portions are spaced from one another along the circumference of the structure by a thinner portion arranged in between.

[0072] In some implementations, at least one of the thicker portions and/or at least one of the

thinner portions extends along, e.g., in parallel, to the axis of the structure. In some implementations, the axis is surrounded by a circumference of the structure at which the thinner portions and the thicker portions are alternately arranged. E.g., the circumference may be defined as a perimeter extending along an outer surface of the lateral wall. In some implementations, the axis extends through the opening and/or the front wall. In some implementations, the axis extends in a direction in which the SD part is insertable into the ear canal.

[0073] In some implementations, the mechanical element is configured to be connected to the SD part at a side of the SD part facing an inner region of the ear canal, in particular the tympanic membrane, when the SD part is inserted into the ear canal. In some implementations, the opening of the cavity enclosed by the structure of the mechanical element faces away from the inner region of the ear canal.

[0074] In some implementations, the mechanical element comprises a connector section at which the element is connectable to the SD part. In some implementations, the connector section has a thickness larger than the thickness of the thinner portions. In some implementations, the connector section has a thickness larger than the thickness of the thicker portions arranged around the axis of the structure.

[0075] In some implementations, the element is formed by injection moulding. In some implementations, the element is formed by injection molding of liquid silicone rubber (LSR).

[0076] In some implementations, one or more reinforcement parts made out of a different material than the structure are arranged on the outside and/or on the inside of the structure. In some implementations, at least one reinforcement part is arranged within the cavity to internally support the structure.

[0077] In some implementations, the structure comprises at least two thinner portions which are angled relative to one another. In an embodiment, a direction in which one of the at least two thinner portions vibrates may thus be angled relative to a direction in which another one of the at least two thinner portions vibrates when sound is transmitted through the at least two thinner portions. To illustrate, a first virtual plane may be defined as a plane extending through a perimeter of one of the at least two thinner portions, and a second virtual plane may be defined as a plane extending through a perimeter of another one of the at least two thinner portions, wherein the first virtual plane and the second virtual plane are angled relative to one another. E.g., a normal vector of the first virtual plane may be angled relative to a normal vector of the second virtual plane.

[0078] In some implementations, one or more of the thinner portions have a planar shape. In some implementations, one or more of the thinner portions have a curved shape. In some implementations, one or more of the thicker portions protrude from at least two of the thinner portions at a region of the structure at which the at least two thinner portions are angled relative to one another, e.g., at a corner of the structure.

[0079] In some implementations, the lateral wall surrounds the cavity, e.g., along a circumference of the lateral wall. The lateral wall and/or the front wall may comprise an inner surface delimiting the cavity and an outer surface opposing the inner surface. In some implementations, the axis of the structure is an axis of the lateral wall, e.g., an axis surrounded by the lateral wall and/or an axis extending along the lateral wall in a longitudinal direction. In some implementations, the axis extends through the cavity surrounded by the lateral wall between the opening, in particular a center of the opening, and the front wall, in particular a center of the front wall.

[0080] In some implementations, the front wall comprises a thinner portion. In some implementations, the front wall comprises a thicker portion. In some implementations, the front wall is planar. In some implementations, the front wall is curved. In some implementations, the front wall is curved toward the cavity.

[0081] In some implementations, the lateral wall comprises a portion in which the lateral wall tapers toward the front end. In some implementations, the lateral wall comprises a portion in which a cross section of the lateral wall remains substantially constant, in particular along a direction of

extension in parallel to a central axis.

[0082] In some implementations, the thinner portions are predominantly configured to provide for a transmission of sound. For instance, an area of the structure which is predominantly configured for sound transmission may be provided by one or more thinner portions at the lateral wall and/or by one or more thinner portions at the front wall. In some implementations, one or more of the thicker portions may also be configured for sound transmission, e.g., to a lesser extent than the thinner portions. A sound transmissible area of the structure may thus be provided by one or more thinner portions and/or one or more thicker portions.

[0083] In some implementations, the lateral wall comprises one or more of the thinner portions from which one or more of the thicker portions protrude at an outer surface and/or at an inner surface of the lateral wall. E.g., one or more of the thicker portions may protrude from the thinner portions at a corner region of the lateral wall at which the thinner portions are angled relative to one another and/or one or more of thicker portions may protrude from the thinner portions at a continuous region of the lateral wall at which the thinner portions are continuously joined.

[0084] In some implementations, at least one of the thinner portions of the lateral wall has a different thickness as compared to the thickness of at least one of the thinner portions of the front wall. In some implementations, at least one of the thinner portions of the lateral wall and at least one of the thinner portions of the front wall have an equal thickness.

[0085] In some implementations, a thicker portion leading to the opening may be configured to provide for an attachment of the element to the transducer or sound tube. To illustrate, the rigidity of the structure may be enhanced at the opening by the thicker portion leading to the opening such that a stable attachment can be achieved, e.g., without risking damaging of the structure. In some implementations, the thicker portion leading to the opening extends around a circumference of the lateral wall. In particular, when the thicker portion leading to the opening extends around a circumference of the lateral wall, a uniform stability of the attachment and/or a minimum risk of damaging the structure may be realized.

[0086] In some implementations, one or more of the stiffeners are formed by one or more of the thicker portions of the lateral wall. To illustrate, one or more thicker portions at the lateral wall may constitute stiffeners, in particular to stabilize the lateral wall with regard to a lower stability of the one or more of the thinner portions at the lateral wall as compared to the thicker portions.

[0087] In some implementations, one or more of the stiffeners are formed such that the structure is configured to contact an ear canal wall at the stiffeners when connected to the electroacoustic transducer or sound tube, and when inserted into an ear canal. The stiffeners may thus provide a spacing of the one or more thinner portions at the lateral wall from the surrounding environment, e.g., the ear canal wall, in particular to ensure that vibrations of the thinner portions, e.g., during a sound transmission, are not hindered by the surrounding environment.

[0088] In some implementations, the structure has a cross section approximating a polygon with edges joined at corners, wherein the corners are formed by at least part of the thicker portions, e.g., the stiffeners, and the edges comprise at least part of the thinner portions. E.g., the edges may be formed by at least part of the thinner portions and/or the edges may comprise at least one thicker portion in addition to one or more thinner portions. In some implementations, the polygon is equilateral such that the edges have an equal length. In some implementations, the polygon is equiangular such that the angles at the corners are equal. In some implementations, the angles at two or more of the corners are different. In some implementations, one or more of the edges are curved. In an embodiment, one or more of the edges are planar.

[0089] In some implementations, at least one reinforcement part made out of a different material than the structure is arranged within the cavity to internally support the structure. In some implementations, the reinforcement part is arranged within the cavity to internally support the structure at a position of one or more of the thicker portions. In an embodiment, the reinforcement part is configured as a structure extending between two supported thicker portions. In some

implementations, the reinforcement part has a planar structure. In an embodiment, the reinforcement part has a rectangular shape. In some implementations, the reinforcement part is provided with a cut out facing the front wall and/or a cut out facing the opening. E.g., the cut out may be a circular sector cut out.

[0090] In some implementations, the structure is made of a single material or multiple materials, e.g. silicone rubber and/or another polymer with Young's modulus less than 10 MPa, in particular less than 5 MPa, e.g., less than 3 MPa, and/or being provided with a coating repellent to cerumen.

[0091] In some implementations, a length of the thinner portions of the structure is at least 2 mm and/or at most 10 mm. In some implementations, a diameter of the structure, in particular of the front wall, is at most 7 mm. In some implementations, a thickness of at least one of the thinner portions is 0.05 mm or below, at least for the thinnest parts.

[0092] In some implementations, the element further comprises at least one dome configured for sealing against an ear canal, wherein the at least one dome is integrally formed with the element.

[0093] In some implementations, the structure comprises rounded edges between the thinner portions to avoid irritation or wearing comfort issues when a transducer element with the protection system is worn in the ear, wherein the radius of the rounded edges may be less than 0.5 mm but greater than 0.05 mm. In some implementations, the rounded edges are provided as thicker portions.

[0094] In some implementations, an earpiece and/or an ITE part may comprise the SD part and the mechanical element. The earpiece and/or an ITE part may be configured for at least partial insertion into the ear canal. The SD part may comprise a shell customized to a shape of an individual ear canal of the user. As another example, the SD part may comprise a flexible member, e.g., a dome, which can conform its shape to the shape of the individual ear canal. In some implementations, the electroacoustic transducer is included in the SD part. In some implementations, the hearing device comprises a housing configured to be worn behind an ear of a user. In some implementations, the electroacoustic transducer is included in the housing configured to be worn behind the ear.

[0095] In some implementations, the element comprises one or more recesses arranged along an inner surface of the element, wherein the one or more recesses are configured to engage a corresponding number of protrusions on an outer surface of the SD part. In this way, the element may be connectable to the SD part. E.g., the recess and the protrusion may be ring-shaped. E.g., the protrusion may extend around an outer circumference of the electroacoustic transducer or the sound tube. E.g., the element may comprise a connector section comprising the recess.

[0096] In some implementations, the element may be connectable to the SD part by means of screwing and/or clamping. This may allow for exchanging the protective element by the user.

[0097] In some implementations, the shape of the structure may be such that the corners of the structure protect the thinner portions to be in contact with the ear canal wall so that their vibration is not hindered.

[0098] In some implementations, the element may comprise a 2K overmolded part, in particular having a hard core and a softer portion, wherein the core may serve for pressure equalization.

[0099] Different types of a hearing device can be distinguished by a position at which they are worn at the ear. Some hearing devices, such as behind-the-ear (BTE) hearing aids and receiver-in-the-canal (RIC) hearing aids, typically comprise an earpiece configured to be at least partially inserted into an ear canal of the ear, and an additional housing configured to be worn at a wearing position outside the ear canal, in particular behind the ear of the user. Some other hearing devices, as for instance earbuds, earphones, hearables, earplugs, in-the-ear (ITE) hearing aids, invisible-in-the-canal (IIC) hearing aids, and completely-in-the-canal (CIC) hearing aids, commonly comprise such an earpiece to be worn at least partially inside the ear canal without an additional housing for wearing at the different ear position.

[0100] FIG. 1 illustrates an exemplary hearing device **10** according to some embodiments of the

present disclosure. The present embodiments aim at providing sound transmission through a mechanical element **12** that can be connected to a sound delivery (SD) part **20** of hearing device **10** comprising a sound tube **16**. The SD part **20** may include an electroacoustic transducer and/or sound tube **16** may be acoustically coupled to an electroacoustic transducer external from SD part **20**. Mechanical element **12** may have a hollow shape with a cavity enclosed by a structure **18**. In some examples, the cavity may be entirely closed but for an opening **8** through which the cavity is acoustically coupled to the electroacoustic transducer when element **12** is connected to SD part **20**. The mechanical element **12** can fulfill multiple purposes including a modifying of a frequency response of hearing device **10** and/or providing a thorough barrier against any substances or fluids that can be present in an ear canal of a user.

[0101] In the example illustrated in FIG. **1**, hearing device **10** is implemented as a RIC hearing aid. RIC hearing aid **10** comprises a BTE part **21** configured to be worn at an ear at a wearing position behind the ear, and ITE part **11** configured to be worn at the ear at a wearing position at least partially inside an ear canal of the ear. Hearing device **10** further comprises mechanical element **12** connectable to ITE part **11**.

[0102] ITE part **11** is implemented as an earpiece at least partially insertable into an ear canal. ITE part **11** comprises SD part **20**. SD part **20** comprises an electroacoustic transducer **17** including a housing **14** accommodating electrical components **13**. In the illustrated example, electroacoustic transducer **17** is implemented as a receiver or loudspeaker, wherein electrical components **13** are configured to convert an electrical signal into sound. In particular, electroacoustic transducer **17** may be implemented as a balanced armature transducer. In other examples, electroacoustic transducer **17** may be implemented as a moving coil speaker. In still other examples, electroacoustic transducer **17** may be implemented as a microphone, e.g. an ear-canal microphone, wherein components **13** may be configured to convert sound into an electrical signal. Housing **14** is further provided with a sound tube **16** for emitting the sound generated by components **13** toward an inner region of the ear canal. Electroacoustic transducer **17** can thus be acoustically coupled to an inner region of the ear canal when SD part **20** is at least partially inserted into the ear canal in order to emit sound into the inner region of the ear canal toward the tympanic membrane. SD part **20** may include additional components as may serve a particular implementation, e.g., an ear canal microphone, a physiological sensor, a processor, and/or the like, which may be accommodated in housing **14**.

[0103] ITE part **11** further comprises a sealing member **15** adapted to contact an ear canal wall when ITE part **11** is at least partially inserted into the ear canal. In the illustrated example, sealing member **15** is provided at SD part **20**. In other examples, sealing member **15** may be integrally formed with mechanical element **12**. Sealing member **15** is implemented as a flexible member configured to conform to the shape of the ear canal wall. For instance, the flexible member may have a shape of a dome. In other examples, sealing member **15** may be provided as a shell having a shape customized to an individual ear canal.

[0104] ITE part **11** further comprises mechanical element **12**. Mechanical element **12** is configured to be mechanically connected to SD part **20**. When connected, as illustrated in FIG. **1**, mechanical element **12** can be inserted into the ear canal along with SD part **20**. Mechanical element **12** comprises structure **18** enclosing a cavity with opening **19** facing the SD part **20**. Thus, element **12** can be configured to be acoustically coupled to electroacoustic transducer **17** via opening **19** facing the SD part **20**. In some implementations, as illustrated, mechanical element **12** is connected to electroacoustic transducer **17**, e.g., at transducer housing **14**. For instance, when mechanical element **12** is connected to electroacoustic transducer **17**, sound tube **16** may be received by element **12** through opening **19** in order to provide for the acoustical coupling to electroacoustic transducer **17**. In other examples, mechanical element **12** may be connected to at least another part and/or another component of SD part **20**, e.g., to sealing member **15**, in a way that element **12** is acoustically coupled to electroacoustic transducer **17** through opening **19**.

[0105] BTE part **21** comprises a BTE housing **24** configured to be worn behind the ear. BTE part **21** and ITE part **11** are interconnected by a cable **22**. A processor **26** accommodated in BTE housing **24** is communicatively coupled to electroacoustic transducer **17** via cable **19** and a cable connector **29** provided at BTE housing **24**. Processor **26** can thus be configured to provide an audio signal to electroacoustic transducer **17** based on which the sound is generated. In the illustrated example, processor **26** is further operatively connected to an electroacoustic transducer **27** configured to convert sound into an electrical signal, which may be implemented by a microphone and/or a microphone array, and a user interface **28**, for instance a switch. BTE part **21** further includes a battery **23** as a power source for the above described components.

[0106] In some other examples, a hearing device may be implemented as BTE hearing aid. A BTE part of the BTE hearing aid may correspond to BTE part **21** described above, wherein the BTE part of the BTE hearing aid further comprises electroacoustic transducer **17** for converting an electrical signal into sound. An SD part of the BTE hearing aid may comprise a sound tube through which the sound generated by electroacoustic transducer **17** can be delivered into the ear canal. An ITE part of the BTE hearing aid may thus comprise mechanical element **12** connected to such an SD part. When mechanical element **12** is connected to the SD part of the BTE hearing aid, mechanical element **12** can thus be acoustically coupled to electroacoustic transducer **17** via the sound tube. For instance, mechanical element **12** may be connected to the sound tube such that a sound outlet of the sound tube may be received by element **12** through opening **19** in order to provide for the acoustical coupling to electroacoustic transducer **17**. In other examples, mechanical element **12** may be connected to at least another part and/or another component of the SD part of the BTE hearing aid.

[0107] In some other examples, a hearing device may be implemented as an ITE hearing aid, an IIC hearing aid, a CIC hearing aid, or earbud, which may consist of an ITE part or earpiece configured to be at least partially inserted into an ear canal and mechanical element **12**. The ITE part may then comprise, in addition to the components of ITE part **11** described above, one or more of the components of BTE part **21** described above.

[0108] FIG. **2** illustrates an exemplary ITE part **51** comprising an SD part **50** and a mechanical element **62** of a hearing device according to some embodiments of the present disclosure. The hearing device may consist of ITE part **51**, or may comprise further components such as a BTE part. E.g., a RIC hearing aid may comprise ITE part **51** and a BTE part. SD part **50** comprises an electroacoustic transducer **57** implemented as a receiver. Receiver **57** comprises a housing **54** accommodating electrical components **53**. A sound outlet of receiver **57** is provided as a sound tube **58** formed in housing **54**. Sound tube **58** is implemented as a spout extending in a medial direction, in particular toward the tympanic membrane, when ITE part **51** is at least partially inserted into the ear canal. Sound tube **58** has an axis **59** in which direction sound generated by electroacoustic transducer **57** is propagating. Axis **59** extends in a longitudinal direction of sound tube **58**. Axis **59** extends through a center of sound tube **58** and may also be referred to as a central axis.

[0109] Mechanical element **62** can be connected to SD part **50**. In some implementations, mechanical element **62** can be mounted to electroacoustic transducer **57**, e.g., sound tube **58** and/or another portion of housing **54**. For instance, the mounting may be achieved by a recess **65**, e.g. a circular recess **65**, along an inner surface of mechanical element **62** which engages with a corresponding protrusion **55**, e.g. a ring-shaped protrusion **55**, on an outer circumference of sound tube **58**.

[0110] Mechanical element **62** comprises a structure **61** enclosing a cavity **64** with an opening **63**. When mechanical element **62** is connected to SD part **50**, opening **63** faces SD part **50** such that cavity **64** is acoustically coupled to electroacoustic transducer **57**. In some implementations, as illustrated, electroacoustic transducer **57** and/or sound tube **58** may extend into cavity **64** through opening **63**. A portion of electroacoustic transducer **57** and/or sound tube **58** extending into cavity **67** may constitute a reinforcement part to internally support structure **61**. In other implementations,

electroacoustic transducer **57** and/or sound tube **58** may be provided external from cavity **64** at a distance from opening **63**.

[0111] Structure **61** comprises a front wall **69** at a front end opposing a rear end at which opening **63** is provided, and a lateral wall **68** extending between front wall **69** and opening **63**. Lateral wall **68** surrounds cavity **64**. The front end is a medial end, i.e., front wall **69** faces the tympanic membrane inside the ear canal when ITE part **51** is at least partially inserted. The rear end is a lateral end such that opening **63** faces the ear canal entrance.

[0112] Structure **61** comprises a first section **66** extending between a front end of sound tube **58** and the front end of structure **61**. First section **66** comprises front wall **69** and a portion of lateral wall **68** leading to front wall **69**. The portion of lateral wall **68** comprised in first section **66** surrounds a volume portion of cavity **64** which remains hollow, in particular empty, when mechanical element **62** is connected to SD part **50**. First section **66** comprises thinner portions **71** and thicker portions **72** alternately arranged around an axis **79** of structure **61**. Axis **79** extends in a longitudinal direction of first section **66** of structure **61**, in particular along lateral wall **68**, and may also be referred to as a longitudinal axis. In the illustrated example, axis **79** extends through a center of first section **66** and may also be referred to as a central axis. When mechanical element **62** is connected to SD part **50**, axis **79** of structure **61** is aligned within 20° , e.g., 10° , relative to axis **59** of sound tube **58**. An angle enclosed between axis **79** of structure **61** and axis **59** of sound tube **58** may thus not exceed 20° . In the illustrated example, axis **79** of structure **61** is substantially overlapping and/or concentric with axis **59** of sound tube **58**. In other examples, axis **79** of structure **61** may be positioned at a distance from axis **59** of sound tube **58**, e.g., shifted in parallel to axis **59** of sound tube **58**.

[0113] Thinner portions **71** and thicker portions **72** alternately arranged alternately arranged around a circumference of structure **61**, in particular around a circumference of lateral wall **68**. In particular, each of thicker portions **72** may be arranged between two of thinner portions **71**. Neighbouring thinner portions **71** may thus be circumferentially spaced by thicker portion **72** arranged in between. In some implementations, thinner portions **71** and thicker portions **72** extend in a longitudinal direction in parallel to axis **79** of structure **61**. In particular, central axis **79** may be surrounded by the portion of lateral wall **68** comprised in first section **66**. In some implementations, thinner portions **71** and/or thicker portions **72** lead to front wall **69**. In some implementations, as illustrated, front wall **69** is a thinner portion **71**. On other implementations, front wall **69** may be implemented as a thicker portion **72**.

[0114] First section **66** of structure **61** can be exploited to transmit sound emitted by electroacoustic transducer **57** from cavity **64** to an exterior of structure **61**, in particular an inner region of the ear canal when inserted into the ear canal. First section **66** may thus also be denoted as a membrane or a membrane section. In particular, a transmission of sound through structure **61** can be more effective at thinner portions **71** and a rigidity of structure **61** can be enhanced by thicker portions **72**. To illustrate, thinner portions **71** may be predominantly employed to transmit sound and thicker portions **72** may predominantly provide for a different functionality, e.g., stabilizing structure **61** and/or maintaining a desired shape of mechanical element **62** also during sound transmission. In particular, thinner portions **71** may more easily be stimulated to vibrate during sound transmission, e.g., with a smaller mechanical damping and/or a larger vibration amplitude, as compared to thicker portions **72** due to a smaller mass of the thinner portions **71**.

[0115] Compared to flat membranes known in the art, a much larger surface can be realized by structure **61**, resulting in improved sound transmission. A further advantage of embodiments described herein is the possibility to adapt the geometry of the acoustically active elements of structure **61**. For example, a frequency response of the hearing device may be altered according to particular specifications and/or an acoustic pressure inside the ear canal may be increased in a particular frequency range and/or at least one resonance frequency may be shifted and/or added in the frequency response and/or a resonance peak may be broadened and/or a cutoff frequency may

be altered and/or an improved acoustic signal output may be provided in a certain frequency band. The geometry membrane section **66** may be optimized to match a desired acoustic performance. In some implementations, a mesh or a foam with open cells may be further included inside cavity **64** to enhance the performance.

[0116] Structure **61** comprises a second section **67** comprising a portion of lateral wall **68** extending between opening **63** and the front end of sound tube **58**. The portion of lateral wall **68** comprised in second section **67** surrounds a volume portion of cavity **64** which is filled by a portion of SD part **50** when element **62** is connected to SD part **50**. Second section **67** comprises a thicker portion **73** providing for a sufficient robustness for mounting mechanical element **62** to SD part **50**, in particular electroacoustic transducer **57**. Second section **67** may thus also be denoted as a connector or a mounting section. Thicker portion **73** may be denoted as a connector or mounting portion. Thicker portion **73** may extend around a circumference of second section **67**, e.g., with a homogenous thickness. In some implementations, thicker portion **73** may be provided with a larger thickness as compared to thicker portions **72** included in first section **66**. An inner surface **77** of thicker portion **73** delimiting cavity **64** at opening **63** may have a shape corresponding to the shape of the portion of electroacoustic transducer **57** and/or sound tube **58**. Protrusion **55** may be formed in second section **67**.

[0117] Mechanical element **62** may be adapted for different ear canal geometries, e.g. small or large ear canals or circular and elliptical ones. In some implementations, mechanical element **62** can be combined together with a sealing member, e.g., a dome, into one single element. In particular, mechanical element **62** may provide for a mechanical coupling to electroacoustic transducer **57**, e.g., sound tube **58** and/or another portion of housing **54**, as it is implemented with prior art domes.

[0118] By choosing an appropriate material, e.g., silicone rubber, of mechanical element **62** and/or by applying a coating, an adhesion of cerumen to a surface of mechanical element **62** can be reduced such that a cleaning of mechanical element **62** can be minimized. Further, due to a smooth surface, it is much easier to clean a mechanical element **62** with a smooth surface than a puncture-sensitive membrane. In some implementations, a material of mechanical element **62** is selected to be biocompatible due to a potential contact with skin. An increased reliability of mechanical element **62** may reduce the need for transducer servicing and less spare parts, e.g. wax filters, may be needed.

[0119] In an illustrative example, an effective length of structure **61**, in particular of thinner portions **71** provided in membrane section **66**, may be 2 mm to 10 mm, e.g. about 5 mm. A diameter of structure **61**, in particular of membrane section **66**, may be 1 mm to 7 mm. A thickness of thinner portions **71** in membrane section **66** may be less than 0.15 mm. For instance, thinner portions **71** may at least partially be provided with a thickness of 0.05 mm or below, e.g., 0.02 mm for the thinnest regions. Improvements in a processing of liquid silicone rubbers in the recent years opened the possibility to create parts with membranes as thin as required for the present application, yet stable enough to sustain the mechanical loads.

[0120] FIGS. 3A and 3B schematically illustrate an arrangement in which ITE part **51** is at least partially inserted into an ear canal **80**. Ear canal **80** is delimited by an ear canal wall **81** leading toward a tympanic membrane **82**. ITE part **51** further comprises a sealing member **79** configured to provide for an acoustical seal against ear canal wall **81**. For instance, sealing member **79** may be implemented as a flexible member, as illustrated, e.g., a dome, or a custom shell. In the illustrated example, sealing member **79** is included in SD part **50**. E.g., sealing member **79** may be attached to electroacoustic transducer **57**, in particular at spout **58** and/or at another part of the transducer housing. In other examples, sealing member **79** may be integrally formed with mechanical element **62**. An acoustical seal with the ear canal wall may thus be provided at a portion of ITE part **51** contacting the ear canal wall. The acoustical seal may at least partially block ambient sound from entering the inner region of ear canal **80** and/or the sound waves generated by electroacoustic

transducer **57** from entering an ambient environment outside ear canal **80**. An inner region of ear canal **80** may be defined as a region between tympanic membrane **82** and ITE part **51** up to a position at ear canal wall **81** at which the acoustical seal by sealing member **79** is provided, when ITE part **51** is at least partially inserted into ear canal **80**.

[0121] In a first arrangement illustrated in FIG. 3A, only SD part **50**, from which mechanical element **62** is disconnected, is inserted into ear canal **80**. The inner region of ear canal **80** has a volume **83**. Volume **83** may be denoted as an ear canal volume.

[0122] In a second arrangement illustrated in FIG. 3B, SD part **50** and mechanical element **62** connected to SD part **50** is inserted into ear canal **80**. Mechanical element **62** comprises structure **61** enclosing cavity **64**, in particular an empty portion of cavity **64**, having a volume **84** extending in front of spout **84**. Volume **84** may thus be denoted as an extended spout volume. Accordingly, in the second arrangement, the inner region of ear canal **80** has a volume **85** which, when compared to volume **83** in the first arrangement, is reduced by extended spout volume **84**. Volume **85** may thus be denoted as a residual ear canal volume. As a result, structure **61** of mechanical element **62** divides ear canal volume **83** into two separate volumes, the extended spout volume **84** and the residual ear canal volume **85**.

[0123] Designing mechanical element **62** comprising structure **61** in a way as described above and below can provide for an acoustic pressure inside extended spout volume **84** exceeding an acoustic pressure within residual ear canal volume **85**. To this end, structure **61** may be provided such that volume **84** of the enclosed cavity is minimized and a circumference along which the thinner portions and thicker portions are alternately arranged is maximized. Structure **61** can thus be configured to provide for, when element **62** is connected to the SD part, an acoustic pressure inside the cavity enclosed by structure **61** which is exceeding an acoustic pressure outside the cavity inside ear canal.

[0124] FIGS. 4A-4D schematically illustrate further exemplary ITE parts **101**, **129**, **139**, **149** comprising an SD part **100**, **120** and a mechanical element **112**, **122**, **132**, **142** of a hearing device according to some embodiments of the present disclosure. SD part **100**, **120** comprises a custom shell **105** having a shape at least partially customized to ear canal **80**, in particular to the ear-canal of an individual user. Custom shell **105** can thus constitute a sealing member forming an acoustical seal with ear canal wall **81**. To fabricate custom shell **105**, a user-specific ear canal geometry may be initially determined, e.g., by taking impressions of the user's ear canal. Subsequently, shell **105** can be formed from a resin by taking into account the predetermined individual ear canal geometry, e.g., in a resin based 3D printing technology, such as digital light processing (DLP) or another stereolithography (SLA) process. As illustrated, custom shell **105** may be open at a rear end, which may also be referred to as a lateral end. The opening may also be referred to as a lateral opening. The opening may be covered by a faceplate **108** mountable to the rear end.

[0125] SD part **100** further comprises an electroacoustic transducer **107** accommodated in an inner volume enclosed by custom shell **105**. Electroacoustic transducer **107** is implemented as a receiver. Custom shell **105** further comprises an opening **106** at a front end, which may also be referred to as a medial end. Opening **106** may also be referred to as a medial opening. Medial opening **106** is intended to provide for a functionality to deliver sound generated by electroacoustic transducer **107** into ear canal **80**, in particular into inner region **83**, **85**.

[0126] In the embodiment illustrated in FIG. 4A, mechanical element **112** is connected to electroacoustic transducer **107** such that it extends from the inner volume enclosed by custom shell **105** through medial opening **106** to the outer volume surrounding custom shell **105**. When ITE part **101** is at least partially inserted into ear canal **80**, mechanical element **112** thus extends into inner region **83**, **85**. Mechanical element **112** has a structure **111** including a first section **116** comprising thinner portions **71** and thicker portions **72** alternately arranged around axis **79** of structure **111**. Structure **111** includes a second section **113** at which structure **111** is mounted to electroacoustic transducer **107**. Second section **113** is provided with a circumferential projection **115**, e.g., at a rear

end of structure **111**. Circumferential projection **115** contacts an inner surface of custom shell **105**. E.g., circumferential projection **115** may be elastic to conform to the inner surface. Circumferential projection **115** may provide for an acoustical seal inside the inner volume of custom shell **105** and/or a fixation of mechanical element **112** relative to electroacoustic transducer **107** and/or a fixation electroacoustic transducer **107** inside shell **105**. Thus, mechanical element **112** may provide for sound transmission through thinner portions **71** included in first section **116** which may be located in the inner volume of custom shell **105** and/or outside the inner volume. In the illustrated example, medial opening **106** of shell **105** is significantly wider than mechanical element **112**. Thus, mechanical element **112** may provide for sound transmission through thinner portions **71** included in first section **116** which may be located in the inner volume of custom shell **105** and/or outside the inner volume.

[0127] In the embodiment illustrated in FIG. 4B, ITE part **129** comprises a mechanical element **122** which is connected to custom shell **105** of an SD part **120**, e.g., at the position of medial opening **106**. In this way, the cavity enclosed by a structure **121** of mechanical element **122** can be acoustically coupled to electroacoustic transducer **107** through the opening of the cavity when mechanical element **122** is connected to custom shell **105**. E.g., a click-on interface may be provided to releasably attach and/or detach mechanical element **122** to custom shell **105**. As illustrated, structure **121** comprises a first section **126** in which thinner portions **71** and thicker portions **72** are alternately arranged around axis **79** of structure **121**. In the example, first section **126** of structure **121** has a dome shape, as compared to a cylindrical shape of first section **116** of structure **111** illustrated in FIG. 4A. Structure **121** includes, in addition to first section **126**, a second section **123**. Second section **123** comprises a retention structure **128**. Retention structure **128** may provide for a retention of mechanical element **122** inside custom shell **105** and/or may also provide for a fixation of electroacoustic transducer **107**. Retention structure **128** is at least partially arranged in the inner volume enclosed by custom shell **105**, e.g., at a position close to medial opening **106**. As illustrated, the sound port of electroacoustic transducer **107** may then be arranged outside the inner volume enclosed by custom shell **105** inside the cavity enclosed by structure **121**. E.g., electroacoustic transducer **107** may extend through medial opening **106** or may be arranged in front of medial opening **106**, e.g., at the medial side of shell **105**. In some implementations, a support structure **127** may be arranged within first section **126** and/or second section **123** to suspend electroacoustic transducer **107**.

[0128] In the embodiment illustrated in FIG. 4C, ITE part **139** comprises mechanical element **132** with a structure **131** including, in addition to first section **126**, a second section **133**. Second section **133** comprises a first portion **134** by which mechanical element **132** is connected to electroacoustic transducer **107**. First portion **134** has an opening leading to a cavity enclosed by structure **131** such that the cavity is acoustically coupled to electroacoustic transducer **107**. Second section **133** comprises a second portion **135** by which mechanical element **132** is connected to custom shell **105** of an SD part **100**, e.g., at an outer surface of shell **105**, for instance by a click-on connection. As illustrated, second portion **135** may extend at least partially around first portion **134**. The cavity enclosed by structure **131** may extend in between the first and second portion **134**, **135**. In the illustrated example, second portion **135** at least partially contacts ear canal wall **81**. In this way, in acoustic seal of ITE part **139** may be enhanced.

[0129] ITE part **149** illustrated in FIG. 4D substantially corresponds to ITE part **139** shown in FIG. 4C, with the exception that a mechanical element **142** comprises a structure **141** having a first section **146** in which thinner portions **71** and thicker portions **72** are alternately arranged around axis **79** of structure **111**, wherein first section **146** comprises a cylindrically shaped part **148**. In this way, first section **146** can be more radially spaced from the ear canal wall **81** as compared to dome shaped first section **126** illustrated in FIG. 4C. A radial sound transmission through first section **146** into ear canal **80** may thus be more efficient and/or a volume of the cavity enclosed by first section **146** may thus be advantageously reduced.

[0130] FIGS. 5A-5F schematically illustrate, in a cross sectional view, exemplary implementations of a section **91-98** of structure **18, 61, 111, 121, 131, 141** of mechanical element **12, 62, 112, 122, 132, 142** in which thinner portions **71** and thicker portions **72** are alternately arranged around an axis, in particular around a circumference of section **91-98**. For instance, section **91-98** may be implemented as first section **66**, or the membrane section, of structure **61**. As illustrated, section **91-98** of structure **18, 61** comprises edges joined at corners, wherein the edges are provided as thinner portions **71** and the corners are provided as thicker portions **72**. In some implementations, as illustrated in FIGS. 5B and 5E, one or more thicker portions **72** may also be provided at one or more edges. In particular, each of thicker portions **72** is arranged between two of thinner portions **71**. In particular, each of thicker portions **72** is protruding at an inner and/or outer surface of section **91-98** of structure **18, 61**.

[0131] In the illustrated examples, section **91-98** of structure **18, 61** has a cross section approximating a polygon. Sections **91-93** illustrated in FIGS. 5A-5C are shaped as a convex polygon. In particular, section **91** illustrated in FIG. 5A has a triangular cross section. An angle **88** spanned between two of thinner portions **71** neighbouring each other, in particular two of thinner portions **71** joined by one thicker portion **72** arranged in between, at an outer surface of the circumference of section **91** is larger than 270° . Section **92** illustrated in FIG. 4B has a rectangular cross section. Accordingly, angle **88** spanned between two neighboring thinner portions **71** at the circumference is 270° . Section **93** illustrated in FIG. 5C has a cross section of a pentagon. Accordingly, angle **88** spanned between two neighboring thinner portions **71** at the circumference is at most 240° .

[0132] Sections **94-98** illustrated in FIGS. 5A-5C are shaped as a concave polygon. Such a shape of section **94-98** of the structure may be exploited to minimize a volume of the cavity enclosed by the structure and to maximize a surface of a circumference along which thinner portions **71** and thicker portions **72** are alternately arranged. In particular, section **94-98** of structure **18, 61** has a cross section touching an outer tangent curve **87, 89** of a larger radius at which at least two of thicker portions **72** are located and an inner tangent curve **86** of a smaller radius at which at least two other of thicker portions **72** are located, wherein at least two of thinner portions **71** extend between outer tangent curve **87, 89** and inner tangent curve **86**. In some examples, as illustrated, outer tangent curve **87, 89** and/or inner tangent curve **86** is elliptical. In particular, as illustrated in FIGS. 5D-5G, outer tangent curve **87, 89** and inner tangent curve **86** may be circular. In the example illustrated in FIG. 5H, only inner tangent curve **86** is circular, wherein outer tangent curve **87, 89** is an ellipse surrounding two focal points.

[0133] Sections **94-98** have a cross section with a plurality of radial projections, wherein each of the projections comprises at least two thinner portions **71**. In particular, sections **94-96** have a star shaped cross section comprising a plurality of circularly arranged projections, in particular jags or lugs or ledges, wherein each jag or lug comprises two thinner portions **71** joined at a radially outer edge formed by a thicker portion **72**. Section **94** illustrated in FIG. 5D has a star-shape with three projections. Two thinner portions **71** which are joined by a thicker portion **72** located at inner tangent curve **86** span angle **88** at an outer surface of the circumference of at most 120° . Section **95** illustrated in FIG. 5E has a star-shape with four projections. Two thinner portions **71** which are joined by a thicker portion **72** located at inner tangent curve **86** span angle **88** of at most 90° . Section **96** illustrated in FIG. 5F has a star-shape with six projections. Two thinner portions **71** which are joined by a thicker portion **72** located at inner tangent curve **86** span angle **88** of at most 60° . Section **97** illustrated in FIG. 5G has a star-shape with five projections. Two thinner portions **71** which are joined by a thicker portion **72** located at inner tangent curve **86** span angle **88** of at most 72° . In the illustrated example, one or more of thicker portions **72** are positioned at a distance from tangent curves **86, 87, 89**, in particular outer tangent curve **87, 89**. In the other illustrated examples of Sections **94-96, 98**, all thicker portions **72** are positioned on tangent curves **86, 87, 89**. Section **98** illustrated in FIG. 5H has another star-shape with five projections.

[0134] Thinner portions **71** may each be defined as a portion, in particular a congruent portion, of the structure not exceeding, i.e. falling below or being equal, a thickness limit. Thus, thinner portion **71** may be thinner than the thickness limit at any location of the structure over which thinner portion **71** extends. Thicker portion **72**, e.g., a thicker portion adjoining the thinner portion **71**, may be defined as a portion exceeding, i.e. not falling below and not being equal, the thickness limit. Thus, thicker portion **72** may be thicker than the thickness limit at any location of the structure over which thicker portion **72** extends. A transition region between thinner portion **71** and thicker portion **72** may be defined as a region at which a thickness of the structure changes from being thinner to being thicker than the thickness limit. Neighboring thinner portions **71** may be defined as two portions not exceeding the thickness limit, wherein a thicker portion **72** exceeding the thickness limit is arranged in between.

[0135] To illustrate, the thickness limit may be selected as a value of 0.15 mm or smaller. Thus, when the thickness limit is selected as a value of 0.14 mm, thinner portion **71** can be defined as portion having a thickness not exceeding 0.14 mm, and thicker portion **72** can be defined as portion having a thickness exceeding 0.14 mm. A transition region between thinner portion **71** and thicker portion **72** can then be defined as a region at which a thickness of the structure changes from below 0.14 mm to above 0.14 mm. When the thickness limit is selected as a value of 0.1 mm, thinner portion **71** can be defined as portion having a thickness not exceeding 0.1 mm, and thicker portion **72** can be defined as portion having a thickness exceeding 0.1 mm. A transition region between thinner portion **71** and thicker portion **72** can then be defined as a region at which a thickness of the structure changes from below 0.1 mm to above 0.1 mm.

[0136] In particular, one or more of thinner portions **71** may have a homogeneous, e.g., constant, thickness not exceeding the thickness limit and/or one or more of thinner portions **71** may have a varying, e.g., increasing and/or decreasing, thickness not exceeding the thickness limit.

Correspondingly, one or more of thicker portions **72** may have a homogeneous, e.g., constant, thickness exceeding the thickness limit and/or one or more of thicker portions **72** may have a varying e.g., increasing and/or decreasing, thickness exceeding the thickness limit. The thickness limit of at least two thinner portions **71** may be selected to be equal or different. Correspondingly, the thickness limit of at least two thicker portions **72**, e.g., thicker portions **72** each adjoining different thinner portions **71**, may be selected to be equal or different.

[0137] FIGS. **6A-6D** schematically illustrate further exemplary mechanical elements **232**, **242**, **252**, **262** configured to be connected to SD part **20**, **50**, **100**, **120**. Mechanical elements **232**, **242**, **252**, **262** comprise a structure **231**, **241**, **261** with thinner portions **71** and thicker portions **72** alternately arranged around axis **79** of structure **231**, **241**, **261**. Structure **231**, **241**, **261** has a cross section shaped as a convex polygon, in particular a hexagon, with edges joined as corners, wherein the corners are formed by thicker portions **72** and the edges comprise thinner portions **71**. In the illustrated example, thinner portions **71** are curved. In particular, thinner portions **71** have a concave curvature at an outer surface of structure **231**, **241**, **261**.

[0138] Mechanical element **232** illustrated in FIG. **6A** comprises a first section **236** with thinner portions **71** and thicker portions **72** alternately arranged around axis **79**, and a second section **233** constituting a mounting section to a sound tube and/or an electroacoustic transducer. Mechanical element **232**, which includes front wall **69** and lateral wall **68** surrounding longitudinal axis **79**, is shaped similar to a cup or beaker. Thicker portions **72** constitute stiffeners **234**, in particular fins, spaced from one another along a circumference of lateral wall **68**. Fins **234** are uniformly distributed around a circumference of lateral wall **68**. E.g., the resulting shape may be similar to a hexalobular external key (e.g. Torx™). Fins **234** improve mechanical stability while thinner portions **71** may predominantly provide for the acoustic performance.

[0139] Mechanical element **242** illustrated in FIG. **6B** comprises a structure **241** constituted by section **236** described above. A respective reinforcement part **245** is arranged on two opposite ones of the fins **234** in the shape of an additional rib on the fins **234**. The reinforcement part **245** is

configured to provide mechanical stability without impact on the acoustic performance. Reinforcement parts **245** on the outer surface of structure **241** may be easily added in a manufacturing process of mechanical element **242**, but they increase the overall size of element **242**.

[0140] Mechanical element **252** illustrated in FIG. **6C** comprises structure **241** described above, wherein a reinforcement part **253** is arranged within the cavity enclosed by structure **241** to internally support two opposite ones of fins **234**. The reinforcement part **253** may be configured as a planar structure extending between the two supported fins **234**. In an exemplary embodiment, a planar structure of reinforcement part **253** may be based on a rectangular shape with a cut-out, e.g. a circular sector, in particular a half circle, cut out respectively on two opposing sides, one facing front wall **69** and the other facing opening **63** of structure **241** opposing front wall **69**.

Reinforcement part **253** is configured to provide mechanical stability without impact on the acoustic performance. The cut-out serves for providing an equal acoustic pressure at different positions inside the cavity. The cut-out does not necessarily have to be circular. Reinforcement part **253** on the inside, as illustrated, may support mainly along one dimension. In other examples, elements providing support in more directions are also conceivable. Reinforcement part **253** may be made out of a different material than structure **241** and inserted in a separate step. In some examples, the electroacoustic transducer and/or sound tube may extend into the cavity, wherein a portion of the electroacoustic transducer and/or the sound tube extending into the cavity may constitute the reinforcement part to internally support the structure.

[0141] Mechanical element **262** illustrated in FIG. **6D** comprises a structure **261** having front wall **69** provided with a rounded portion **266** to prevent irritations in case of contact with the ear canal skin. Rounded portion **266** may be arranged circumferentially around a perimeter of front wall **69**.

[0142] FIGS. **7A**, **7B** schematically illustrate further exemplary mechanical elements **272**, **282** configured to be connected to SD part **20**, **50**, **100**, **120**. Mechanical element **272**, **282** comprises a structure **271**, **281** with thinner portions **71** and thicker portions **72** alternately arranged around axis **79** of structure **271**, **281**. Structure **271**, **281** comprises a first section **276**, which may be implemented corresponding to first section **66**, **116**, **126**, **146**, **236** described above, and a second section **273** configured to mount mechanical element **272**, **282** to SD part **20**, **50**, **100**, **120**.

Structure **271** illustrated in FIG. **7A** is provided with a sealing member **279**, e.g., a dome, for sealing against the ear canal of the user. Sealing member **279** may be integrally formed with structure **271**. As illustrated, sealing member **279** may be provided at second section **273**, e.g., in proximity to first section **276** and/or in proximity to opening **63** and/or in between first section **276** and opening **63**.

[0143] Structure **281** illustrated in FIG. **7B** is provided, in addition to sealing member **279**, with a further sealing member **289**, e.g., a further dome. Sealing member **289** may also be integrally formed with structure **281**. As illustrated, sealing member **289** may be provided at first section **276**, e.g., in proximity to front end **69** and/or between front end **69** and a transition portion between first section **276** and second section **273**. Sealing member **289** may be implemented as an open sealing, e.g., an open dome, to maintain a desired functionality. An open dome is a dome with a relatively large acoustically open cross-section. This implies that a large amount of direct sound can enter through the sealing. For instance, sealing member **289** may be provided with one or more venting channels.

[0144] FIGS. **8A-8D** illustrate a mechanical element **362** according to some embodiments of the present disclosure. FIG. **8A** depicts element **362** in a perspective view, FIG. **8B** in a longitudinal sectional view along a cutting plane XVI shown in FIG. **8A**, FIG. **8C** in a profile cross sectional view along a cutting plane XVII shown in FIG. **8A**, and FIG. **8D** in a profile longitudinal view along a cutting plane XVI. Element **362** comprises a structure **361** with a section **366** in which thinner portions **71** and thicker portions **72** are alternately arranged around axis **79** of structure **361**, in particular along a circumference of structure **361**.

[0145] Thinner portions **71** are angled relative to one another, in particular at an angle smaller than 180°. E.g., Thinner portions **71** may be associated with a respective virtual plane extending through a perimeter of the respective thinner portion **71**, wherein respective normal vectors of the planes are angled relative to one another. Each of thinner portions **71** can thus be configured to vibrate in a different direction, e.g., in the direction of the normal vector, in order to transmit sound. In this way, an effective area of section **366**, which is predominantly configured for sound transmission, can be effectively increased within a restricted space such as inside an ear canal, for instance as compared to a single flat-shaped membrane. E.g., structure **361** may include at least part of the thinner portions **71** in a polyhedric arrangement. An angle at which thinner portions **71** are angled relative to one another may be defined as an angle in between the respective virtual planes extending through the perimeter of the thinner portions **71**.

[0146] In the illustrated example, a front wall **369** at a front end **367** of structure **361** is formed by thinner portion **71**. In other examples, front wall **369** may comprise at least one thicker portion **72**. Front end **367** is opposing a rear end **374** of structure **361** at which opening **63** is provided. A lateral wall **368** extends between front wall **369** at front end **367** and opening **63** at rear end **374**. Lateral wall **368** comprises a plurality of thinner portions **71** surrounding cavity **64**, e.g., as illustrated, four thinner portions **71**. In particular, section **92** illustrated in FIG. 5B may be implemented in such a way. Front wall **369** and lateral wall **368** comprise an inner surface delimiting cavity **64** and an outer surface opposing the inner surface. The outer surface of the lateral wall **368** may define a lateral area of section **366**. The outer surface of the front wall **369** may define a bottom of section **366**. As illustrated, section **366** may have a cup-like or beaker-like shape with a bottom provided at front wall **369** and a lateral area provided at lateral wall **368**.

[0147] Axis **79** extends through cavity **64** surrounded by lateral wall **368** between front end **369**, at which front wall **369** is provided, and rear end **374**, at which opening **63** is provided, e.g., between a center of opening **63** and a center of front wall **369**. Cutting plane XVI extends through central axis **79**. Central axis **79** is normal to cutting plane XVII. Lateral wall **368** further comprises thicker portions **72**. At least part of the thicker portions **72** protrude from at least part of thinner portions **71** at the inner surface of lateral wall **368**. In other examples, as described above, lateral wall **368** may also comprise thicker portions **72** protruding from at least part of thinner portions **71** at the outer surface of lateral wall **368**.

[0148] Structure **361** further comprises, in addition to first section **366**, a second section **363** of lateral wall **368**. Second section **363** leads to opening **63** at rear end **374**. Second section **363** is implemented as a mounting section. Structure **361** can thus be configured to be attached to the SD part **20, 50, 100, 120**, e.g., the electroacoustic transducer and/or sound tube, by means of mounting section **363**. A sealing member **379**, implemented as a dome is integrally formed with structure **361**. Sealing member **379** is attached to second section **363** at a distance from rear end **374**.

[0149] A plurality of thicker portions **72** of lateral wall **368** extend between two thinner portions **71** respectively. Thicker portions **72** form stiffeners of section **366** by means of which a stiffness of structure **361** can be increased between thinner portions **71** due to the larger rigidity of stiffener **72**. Stiffeners **72** may extend in parallel to axis **79** in the form of fins. Second section **363** may be thicker than stiffeners **72** to account for an increased stability required for the attachment to SD part **20, 50, 100, 120**.

[0150] As illustrated in FIG. 8C, the thicker portions **72** protrude from thinner portions **71** at a region of the lateral wall **368** at which two respective thinner portions **71** are joined at an angle. The region of the lateral wall **368** at which the two thinner portions **71** are joined at the angle may constitute a corner region of the lateral wall **368**, as illustrated. Thicker portions **72** can then be formed in at least part of the corner regions of the lateral wall **368**. As further illustrated in FIG. 8C, a thickness of structure **361** progressively decreases at thicker portions **72** toward the adjoining thinner portions **71**, in particular in a direction in which the thicker portions **72** lead to adjoining thinner portions **71**. In some examples, also a thickness of the thinner portions **71** may vary. E.g.,

thinner portions **71** may become thicker toward thicker portions **72**. In this way, a smooth transition between the thinner portions **71** and thicker portions **72** may be provided for. In other examples, thinner portions **71** and/or thicker portions **72** may be provided with a homogenous thickness. [0151] As illustrated in FIGS. **8C** and **8D**, thinner portions **71** may have a curved shape. In particular, thinner portion **71** at front end **367** may comprises an inward curvature toward cavity **64**. Thinner portions **71** of lateral wall **368** may comprise an outward curvature away from cavity **64**. In this way, an effective area of structure **361** may be further increased.

[0152] As illustrated in FIGS. **8B** and **8D**, lateral wall **368** comprises a substantially cylindrical part **375** in which a cross section of lateral wall **368** remains constant, e.g., with regard to the direction of extension along longitudinal axis **79**. Lateral wall **368** further comprises a conical part **376** in which a cross section of lateral wall **368** continuously decreases toward front end **369**. Conical part **375** may facilitate insertion of mechanical element **362** into an ear canal and/or provide for an increased stability of structure **361** at a part of structure **361** which is more distant from the attachment to SD part **20, 50, 100, 120**.

[0153] As illustrated in FIG. **8A**, thinner portion **71** at front end **367** substantially extends perpendicular to thinner portions **71** of lateral wall **368**. As illustrated in FIG. **8C**, an angle spanned between two neighbouring thinner portions **71** of lateral wall **368** at an outer circumference of structure **361** is larger than 200°. In particular, the angle spanned between some of thinner portions **71** at the circumference of structure **361**, as illustrated at the top and bottom of FIG. **8C**, is smaller than 270°, and the angle spanned between some other of thinner portions **71** at the circumference, as illustrated at the left and right of FIG. **8C**, is larger than 270°.

[0154] As illustrated in FIG. **8A**, lateral wall **368** of structure **361** has a cross section approximating a polygon with edges joined at corners, wherein the corners are formed by thicker portions **72** and the edges comprise thinner portions **71**. In particular, lateral wall **368** has a cross section approximating a rhombus with four corners formed by respective stiffeners, in particular fins, constituted by thicker portions **72**. A diameter of lateral wall **368** across two opposing ones of corners **72** is greater than a diameter of the lateral wall **368** across the two other opposing corners **72**. In the illustrated example, the edges between the corners **72**, as formed by thinner portions **71**, are slightly convex.

[0155] FIGS. **9A-9D** illustrate a mechanical element **162** according to some embodiments of the present disclosure. FIG. **9A** depicts element **162** in a perspective view, FIG. **9B** in a longitudinal sectional view along a cutting plane XVI shown in FIG. **9A**, FIG. **6C** in a front view, and FIG. **9D** in a cross sectional view along a cutting plane XVII shown in FIG. **9A**. Element **162** comprises a structure **161** with a section **166** in which thinner portions **71** and thicker portions **72** are alternately arranged around a circumference of the structure **161**. In some examples, as illustrated, element **162** consists of section **166**. In other examples, as further described below, the element may comprise at least one additional section. For instance, first section **66** of structure **61** of mechanical element **62** illustrated in FIG. **2** may be implemented corresponding to section **166**.

[0156] Structure **161** comprises a front wall **169** at a front end opposing a rear end at which opening **63** is provided. Structure **61** further comprises a lateral wall **168** surrounding cavity **64** and extending along central axis **79** between front wall **169** and opening **63**. In the illustrated example, front wall **169** is implemented as a thicker portion **72**. Structure **161** may be connected to SD part **20, 50** by an additional mounting means, e.g., a clamp or the like. In particular, the mounting means may comprise an opening having a shape corresponding to the shape of opening **63** to provide for an acoustical coupling of cavity **64** through opening **63** to electroacoustic transducer **17, 54** and/or sound tube **58** included in SD part **20, 50**.

[0157] Structure **161** has a cross section with a plurality of radial projections **178** each comprising at least two thinner portions **71** joined by a radially outer thicker portion **72**. Radial projections **178** are shaped as ledges, wherein thinner portions **71** extend substantially in parallel to one another. Outer thicker portions **72** are substantially flat. Each two neighbouring ledges **178** are joined by a

radially inner thicker portion **72**. In this way, ledges **178** are spaced from another at their radially inner end, wherein the distance between the ledges **178** increases toward their radially outer end. Thicker portions **72** form stiffeners of section **166** by means of which a stiffness of structure **161** can be increased between thinner portions **71** due to the larger rigidity of stiffener **72**. Stiffeners **72** extend along axis **79** in the form of fins.

[0158] In the illustrated example, different projections **178** are equidistantly spaced from one another. Angle **88**, which is spanned between opposing thinner portions **71** of neighboring projections **178**, is at most 60° . In other examples, different projections **178** may have a different spacing. Ledges **178** are arranged around a circle forming a star shaped cross section of structure **61**. In the illustrated example, six ledges **178** are circularly arranged. For instance, section **96** illustrated in FIG. 5F may be implemented as structure **161**.

[0159] Thinner portions **71** have a width **171** between adjoining thicker portions **72**. Radially outer thicker portions **72** have a width **172** between adjoining thinner portions **71**. Radially inner thicker portions **72** have a width **173** between adjoining thinner portions **71**. In the illustrated example, width **171** of thinner portions **71** exceeds width **172**, **173** of adjoining thicker portions **72** by at least a factor of two.

[0160] FIGS. **10A** and **10B** illustrate another mechanical element **182** according to some embodiments of the present disclosure. FIG. **10A** depicts element **182** in a perspective view, and FIG. **10B** in a longitudinal sectional view along a cutting plane XVI shown in FIG. **10A**. Element **182** comprises a structure **181** with a first section **186** implemented corresponding to section **166** described above, and a second section **183**. Structure **161** comprises front wall **169** at the front end opposing the rear end at which opening **63** is provided, and a lateral wall **188** surrounding cavity **64** and extending along central axis **79** between front wall **169** and opening **63**. Second section **183** is a connector or mounting section for mounting mechanical element **62** to SD part **50**, in particular electroacoustic transducer **57** and/or a sound tube. Lateral wall **188** comprises a thicker portion **183** at second section **183** in which at least part of electroacoustic transducer **57** and/or the sound tube is insertable. To this end, an inner surface of thicker portion **183** delimiting cavity **64** at opening **63** may have a shape corresponding to the shape of the portion of electroacoustic transducer **57** and/or sound tube **58**.

[0161] Mechanical element **182** further comprises a sealing member **189** for providing an acoustical seal against the ear canal wall. Sealing member **189** can be integrally formed with mechanical element **182**. To this end, sealing member **189** is formed at an outer circumference of lateral wall **188**. E.g., as illustrated, sealing member **189** may be formed at second section **183**, in particular at a front end of second section **183** at which second section **183** adjoins a rear end of first section. In this way, first section **186** can be effectively positioned inside the ear canal in front of the sealing provided by sealing member **189** to provide for sound transmission. Sealing member **189** is flexible to conform to varying ear canal sizes. In the illustrated example, sealing member **189** is implemented as a dome.

[0162] A venting channel **187** is implemented as a through hole in sealing member **189**, e.g., at a position of the dome-like curvature of sealing member **189** which is facing the inner region of the ear canal and/or which is located in front of a position at which sealing member **189** provides for the acoustical seal. The venting channel can provide for a venting between the inner region of the ear canal and the ambient environment outside the ear canal when mechanical element **182** connected to SD part **20**, **50** is at least partially inserted into the ear canal.

[0163] FIGS. **11A** and **11B** illustrate another mechanical element **192** according to some embodiments of the present disclosure. FIG. **11A** depicts element **192** in a perspective view, and FIG. **11A** in a profile cross sectional view along a cutting plane XVII shown in FIG. **11A**. Element **192** comprises a structure **191** with a first section **196** in which thinner portions **71** and thicker portions **72** are alternately arranged around a circumference of the structure **161**, and a second section **193** which is a connector or mounting section for mounting mechanical element **62** to an

SD part. Structure **191** comprises a front wall **199** at a front end opposing the rear end at which opening **63** is provided, and a lateral wall **198** surrounding cavity **64** and extending along central axis **79** between front wall **199** and opening **63**.

[0164] First section **196** has a cross section with a plurality of radial projections **194** each comprising at least two thinner portions **71** joined by a radially outer thicker portion **72** forming a star shape. Radial projections **194** are shaped as ledges. The cross section of first section **196** touches an outer tangent curve of a larger radius and an inner tangent curve **86** of a smaller radius, wherein the outer tangent curve is elliptical with two focal points, and the inner tangent curve is circular, similar to cross section **98** depicted in FIG. 5H. Accordingly, a radial length of different thinner portions **71** joined by radially outer thicker portions **72** and/or a radial length of different ledges **194** varies. In the illustrated example, six ledges **194** are circularly arranged in the star shape. Two of the ledges **194** opposing each other have a larger radial length than the four remaining ledges.

[0165] Radially outer thicker portions **72** are curved relative to central axis **79**. The curvature decreases toward the front end at which front wall **199** is arranged. Correspondingly, a radial length of thinner portions **71** joined by radially outer thicker portions **72** and/or a radial length of ledges **194** decreases toward the front end. Correspondingly, lateral wall **198** tapers in first section **196** along central axis **79** toward the front end at which front wall **199** is provided.

[0166] In some implementations, mechanical element **192** may be connected to an electroacoustic transducer and/or a sound tube included in a custom shell to provide for an acoustical sealing. In some other implementations, mechanical element **192** may be directly connected to the custom shell, e.g., at an outer surface of the custom shell at which a sound outlet is provided. In some implementations, mechanical element **192** may be connected to an SD part including a sealing member to provide for the acoustical sealing.

[0167] FIGS. **12A-12C** illustrate further mechanical elements **202**, **212**, **222** according to some embodiments of the present disclosure in a profile cross sectional view. Mechanical element **202** illustrated in FIG. **12A** comprises a structure **201** in which four projections **204** are circularly arranged in a star-like manner, similar to circumference **95** illustrated in FIG. 5E. Angle **88** spanned by neighboring thinner portions **71** at an outer face of structure **201** is at most 90° . Projections **204** are substantially ledge-shaped, wherein a distance between opposing thinner portions **71** slightly decreases in a radial direction. In other examples, the mutual distance between opposing thinner portions **71** of the projections may decrease stronger, as schematically illustrated by the jags formed by thinner portions **71** in FIG. 5E. Mechanical element **212** illustrated in FIG. **12B** comprises a structure **211** in which a star-shape is formed by seven projections **214**. Angle **88** is at most 55° . A distance between opposing thinner portions **71** of projections **214** slightly increases in the radial direction. Mechanical element **222** illustrated in FIG. **12C** comprises a structure **221** with a star-shape of eight projections **224**. Angle **88** is 45° or less. A distance between opposing thinner portions **71** of projections **224** increases stronger in the radial direction as compared to projections **214** in FIG. **12B**.

[0168] FIGS. **13A** and **13B** schematically illustrate one of thinner portions **71** during sound transmission. During sound transmission, thinner portions **71** can be excited to vibrate in a membrane like manner, e.g., similar to a substantially flat membrane. Different heights of a deflection of thinner portions **71** during the vibration are illustrated in FIGS. **13** and **13B** as different contour lines **304**. In the example illustrated in FIG. **13A**, a maximum region of the membrane deflection **305** is shifted from the center toward a position at which a width of thinner portion **71** increases. This corresponds to a first vibrational mode of thinner portion **71**. In the example illustrated in FIG. **13B**, two maximum regions of the membrane deflection **315** are spaced from one another, e.g., about a half of a length of thinner portion **71**. This corresponds to a second vibrational mode of thinner portion **71**.

[0169] Generally, a mechanical and/or acoustic behavior of structures including thinner portions **71**

in the above described way can be selected and/or tuned by a large number of parameters of thinner portions **71** including a thickness, shape, size, number of thinner portions **71**, their geometrical arrangement relative to one another, and their position relative to the opening through which they are acoustically coupled to an electroacoustic transducer. E.g., thinner portions **71** can be configured to exhibit a modal behavior. A first vibrational mode is schematically illustrated in FIG. **9** by the single maximum height deflection **305**. A second mode may exhibit two separate areas of a maximum height deflection. A resonance frequency of the different vibrational modes can thus be tuned by the various design parameters of thinner portions **71**. To illustrate, a first mode may be tuned to be around 3 kHz, a second mode around 6 kHz, and a third mode above 7 kHz. Further, sharp resonances may be achieved when geometrical properties of multiple or all thinner portions **71** are equal. A flattening of the resonance peaks may be achieved by providing different geometrical properties of multiple or all thinner portions **71**.

[0170] FIG. **14** illustrates a functional plot of an exemplary frequency response curve measurable in an inner region of an ear canal when a hearing device including an SD part is at least partially inserted into an ear canal. Alternatively, the frequency response may be measured, e.g., in a 2 cc coupling measurement system. A frequency of sound waves is horizontally indicated on an axis of abscissas. The frequency is indicated in units of Hertz. An acoustic output in units of decibel is vertically indicated on an axis of ordinates. A solid curve **333** represents the frequency response with mechanical element **12, 62, 162, 182, 192, 202, 212, 222** disconnected from the SD part. A dashed curve **333** represents the frequency response when the mechanical element is connected to the SD part.

[0171] As illustrated by solid curve **333**, when the mechanical element is disconnected, a first resonance **334** be observed at about 3 kHz, and a second resonance **335** can be observed at about 7 kHz. As illustrated by solid curve **333**, when the mechanical element is disconnected, a first resonance **336** is only slightly shifted to a lower frequency about 3 kHz, whereas a second resonance **337** is shifted more pronounced to a higher frequency of about 7.5 kHz. A broadening of all resonance peaks **336, 337** can be observed as compared to when the mechanical element is disconnected.

[0172] Further, a third resonance **338** can be observed at about 5 kHz between first and second resonance **336, 337**. Third resonance **338** has a broader peak as compared to resonances **336, 337** leading to an overall flattened frequency response curve between 2 and 6 kHz as compared to when the mechanical element is disconnected. In addition, the acoustic output can be increased in this frequency range as compared to when the mechanical element is disconnected. In addition, when the mechanical element is connected, a cutoff frequency above 8 kHz can be observed, lowering high frequency distortions, which may be attributed to the mechanical element acting as a low pass filter.

[0173] The advantageous frequency response illustrated in FIG. **14** can be obtained with a mechanical element in which section **66, 116, 126, 146, 166, 186, 196, 236, 366** of structure **61, 111, 121, 131, 141, 161, 181, 191, 201, 211, 221, 231, 241, 261, 361** is provided such that a volume of the enclosed cavity **64** is minimized and a circumference of the structure at which thinner portions **71** and thicker portions **72** are alternately arranged, in particular a surface of the circumference, is maximized. In some examples, those geometrical properties of the section of the structure may be achieved by a cross section of the structure with projections **168, 188, 198, 204, 214, 224** extending in the radial direction, in particular a star-shaped cross section, as provided, e.g., by structures **161, 181, 191, 201, 211, 221**. More particularly, a structure including at least three projections, e.g., at least six projections **168, 188, 198, 204, 214, 224**, in the radial direction may be employed to achieve a desired behaviour of the frequency response.

[0174] FIGS. **15A-15B** schematically illustrate another exemplary ITE part **1500** comprising a mechanical element **1502** integrally formed with an SD part **1504**. For example, FIG. **15A** shows a side cross-sectional view of ITE part **1500** and FIG. **15B** shows a cross-sectional view to ITE part

1500 taken along line XV-XV of FIG. 15A. Various components shown in FIGS. 15A-15B may be similar to and/or implement any one or more components of the ITE parts described above.

[0175] To illustrate, SD part **1504** may be similar to and/or implement SD part **20, 50, 100, 120**. As shown, SD part **1504** comprises a custom shell **1506** that may be similar to and/or implement custom shell **105** in that custom shell **1506** has a shape at least partially customized to an ear canal (e.g., ear canal **80**), in particular to the ear-canal of an individual user. Custom shell **1506** can thus constitute a sealing member forming an acoustical seal with an ear canal wall (e.g., ear canal wall **81**). As illustrated, custom shell **1506** includes a sound tube **1508** extending therethrough, such as parallel relative to a longitudinal axis **1510** of custom shell **1506**. Sound tube **1508** may be open at a rear end of custom shell **1506**, which may also be referred to as a lateral end **1512**. The opening may also be referred to as a lateral opening. The opening may be covered by a faceplate **1514** that may be similar to and/or implement faceplate **108** such that faceplate **1514** is mountable to lateral end **1512** of custom shell **1506** (e.g., to cover the lateral opening).

[0176] SD part **1504** further comprises an electroacoustic transducer **1516** that may be similar to and/or implement transducer **107**. Transducer **1516** may be accommodated in an inner volume (e.g., sound tube **1508**) enclosed by custom shell **1506**. Sound tube **1508** of custom shell **1506** may further be open at a front end of custom shell **1506**, which may also be referred to as a medial end. The opening may also be referred to as a medial opening **1518**. Medial opening **1518** may allow sound sound generated by transducer **1516** and guided through sound tube **1508** to be delivered to mechanical element **1502**.

[0177] Mechanical element **1502** may be similar to and/or implement any one or more components of mechanical elements **12, 62, 112, 122, 132, 142, 162, 182, 192, 202, 212, 222, 232, 242, 252, 272, 282, 362**. For example, mechanical element **1502** may include a structure **1520** having a first section **1522-1** and a second section **1522-2**. First section **1522-1** includes one or more thinner portions **71** and thicker portions **72** alternatingly arranged around a circumference of first section **1522-1**. In the illustrated example, first section **1522-1** includes a plurality of radial projections **1524** extending outwardly about a longitudinal axis **1526** of structure **1520** (e.g., forming a star shape), the radial projections **1524** comprising thinner portions **71** and thicker portions **72**. In the illustrated example, radial projections **1524** are hollow such as to form a cavity **1528** within first section **1522-1**. In some implementations, radial projections **1524** lead to a front wall **1530** enclosing cavity **1528**. Front wall **1530** may be implemented as a thinner portion **71** and/or a thicker portion **72**.

[0178] First section **1522-1** of mechanical element **1502** is configured to transmit sound emitted by transducer **1516** from cavity **1528** to an exterior of mechanical element **1502**, in particular to an inner region of the ear canal when ITE pat **1500** is inserted into the ear canal. In particular, a transmission of sound through first section **1522-1** may be more effective at thinner portions **71** and a rigidity of first section **1522-1** may be enhanced by thicker portions **72**. To illustrate, thinner portions **71** may be predominantly employed to transmit sound and thicker portions **72** may predominantly provide for a different functionality, e.g., stabilizing the structure and/or maintaining a desired shape of first section **1522-1** of mechanical element **1502** also during sound transmission. In particular, thinner portions **71** may more easily be stimulated to vibrate during sound transmission, e.g., with a smaller mechanical damping and/or a larger vibration amplitude, as compared to thicker portions **72** due to a smaller mass of the thinner portions **71**.

[0179] While first section **1522-1** is shown as having a star shape, first section **1522-1** of mechanical element **1502** may have any additional or alternative shape that may be adapted for different ear canal geometries, e.g. small or large ear canals or circular and elliptical ones. In some implementations, mechanical element **1502** can be combined together with a sealing member, e.g., a dome, into one single element.

[0180] Second section **1522-2** of structure **1520** integrally connects first section **1522-1** with custom shell **1506**, e.g., at the position of medial opening **1518**. Second section **1522-2** further

comprises an opening **1532** extending therethrough from medial opening **1518** of SD part **1504** to cavity **1528** of first section **1522-1** along longitudinal axis **1526** of structure **1520**. In this way, cavity **1528** enclosed by structure **1520** of mechanical element **1502** may be acoustically coupled to transducer **1516** through opening **1532**. Accordingly, sound emitted by transducer **1516** may be delivered through medial opening **1518** of sound tube **1508**, through opening **1532**, to cavity **1528** of first section **1522-1** for transmission through mechanical element **1502**. Second section **1522-2** may have a structure that is configured to maintain the position and/or orientation of first section **1522-1** relative to custom shell **1506**.

[0181] In the illustrated example, second section **1522-2** is configured to align axis **1526** of structure **1520** to substantially overlap and/or be concentric with axis **1510** of sound tube **1508**. Alternatively, second section **1522-2** may be configured to offset axis **1526** of structure **1520** relative to axis **1510** of sound tube **1508**. Moreover, in the illustrated example, structure **1520** and custom shell **1506** each have an elliptical shape such that second section **1522-2** is configured to align a major axis **1534** of the elliptical shape of structure **1520** with a major axis **1536** of the elliptical shape of custom shell **1506** to substantially overlap. Still other suitable configurations and/or orientations for positioning mechanical element **1502** relative to SD part **1504** may be used. [0182] As an illustrative example, FIGS. **16A-16B** schematically illustrate another exemplary ITE part **1600** that may be similar to and/or implement ITE part **1500** in that ITE part **1600** comprises mechanical element **1502** integrally formed with SD part **1504**. For example, FIG. **16A** shows a side cross-sectional view of ITE part **1600** and FIG. **16B** shows a cross-sectional view to ITE part **1600** taken along line XVI-XVI of FIG. **16A**. However, as shown FIG. **16A**, mechanical element **1502** is connected to SD part **1504** such that axis **1526** of structure **1520** is aligned at an angle **1602** relative to axis **1510** of sound tube **1508**. Angle **1602** between axis **1526** of structure **1520** and axis **1510** of sound tube **1508** may be within 20°, e.g., 10°. Additionally or alternatively, axis **1526** of structure **1520** may be positioned at a distance from axis **1510** of sound tube **1508**, e.g., shifted in parallel to axis **1510** of sound tube **1508**. Accordingly, the orientation of axis **1526** of structure **1520** relative to axis **1510** of sound tube **1508** may be adapted based on a geometry of the ear canal of an individual user.

[0183] Additionally alternatively, as shown in FIG. **16B**, mechanical element **1502** may be rotated relative to SD part **1504** about axis **1526** (e.g., clockwise or counterclockwise about axis **1526**). In this way, a major axis **1534** of the elliptical shape of structure **1520** may be rotated and/or offset relative a major axis **1536** of the elliptical shape of custom shell **1506** at an angle **1604**. Angle **1604** between major axis **1534** of structure **1520** and major axis **1536** of custom shell **1506** may be within plus or minus 90°. The orientation of major axis **1534** of structure **1520** relative to major axis **1536** of custom shell **1506** may be adapted based on a geometry of the ear canal of an individual user.

[0184] To fabricate an ITE part **1500**, **1600** having a custom shell **1506**, a user-specific ear canal geometry may be initially determined, e.g., by taking impressions of the user's ear canal. Subsequently, custom shell **1506** and structure **1520** may be formed by taking into account the predetermined individual ear canal geometry using a mold.

[0185] FIGS. **17A-17B** schematically show an illustrative mold **1700** that may be used to form an ITE part **1500**, **1600** having mechanical element **1502** integrally formed with SD part **1504**. For example, FIG. **17A** shows a side cross-sectional view of mold **1700** and FIG. **17B** shows a perspective cross-sectional view of mold **1700** with a first portion **1702** of mold **1700** omitted for illustrative purposes.

[0186] As shown, mold **1700** comprises a first portion **1702** and a second portion **1704**. First portion **1702** includes a first outer shell **1706** having a first interior **1708** shaped to form custom shell **1506** of SD part **1504**. Accordingly, first interior **1708** may be formed based on the impressions of the individual ear canal such that first interior **1708** is customized for the individual ear canal (e.g., for forming custom shell **1506**). In some examples, first outer shell **1706** of first

portion **1702** is used to form an individual custom shell **1506** such that first outer shell **1706** is not reusable after forming the individual custom shell **1506**. In some examples, first outer shell **1706** may be made from a resin material configured to be broken away and/or dissolved from the formed custom shell **1506**. In some examples, first outer shell **1706** may be formed such as by 3D printing, e.g., in a resin-based 3D printing technology, such as digital light processing (DLP) or another stereolithography (SLA) process.

[0187] Second portion **1704** of mold **1700** includes a second outer shell **1710** having a second interior **1712** shaped to form structure **1520** of mechanical element **1502**. Because structure **1520** of mechanical element **1502** may have a standard shape across different ITE parts **1500**, **1600**, second interior **1712** of second outer shell **1710** may have a standard shape that is associated with multiple ITE parts **1500**, **1600**. Such a standard shape may allow second outer shell **1710** to be reused for forming multiple ITE parts **1500**, **1600**. To illustrate, after being connected with a first outer shell **1706** to form a first mechanical element **1502** integrally with a first custom shell **1506**, second outer shell **1710** of mold **1700** may be reconnected with another first outer shell **1706** such as to form another mechanical element **1502** integrally with another custom shell **1506** having a different shape. In some examples, second outer shell **1710** may be made from a reusable rigid material, such as metal and/or plastic (e.g., thermoplastic). In some examples, second outer shell **1710** may be formed such as by 3D printing, e.g., in a resin-based 3D printing technology, such as digital light processing (DLP) or another stereolithography (SLA) process.

[0188] In the illustrated example, a lateral end **1714** of second outer shell **1710** is positioned at a medial end **1716** of first outer shell **1706** such that first interior **1708** is connected with second interior **1712**. In this way, mechanical element **1502** may be integrally formed with custom shell **1506** using mold **1700**. For example, a material (e.g., silicone) may be introduced into first interior **1708** and second interior **1712** of mold **1700** to form mechanical element **1502** integrally with SD part **1504** in the same material. In the illustrated example, second outer shell **1710** is further aligned with first outer shell **1706** such as to align an axis **1526** of a structure **1520** to be formed within second outer shell **1710** with an axis **1510** of a sound tube **1508** to be formed within first outer shell **1706** (e.g., angle **1602** is about 0°). Alternatively, second outer shell **1710** may be offset and/or angled relative to first outer shell **1706** such as to offset an axis **1526** of a structure **1520** to be formed with an axis **1510** of a sound tube **1508** to be formed (e.g., angle **1602** is greater than about 0° and less than about 20°).

[0189] Moreover, in the illustrated example, second outer shell **1710** is further rotationally aligned with first outer shell **1706** such as to align a major axis **1534** of an elliptical shape of a structure **1520** to be formed within second outer shell **1710** with a major axis **1536** of an elliptical shape of custom shell **1506** to be formed within first outer shell **1706** (e.g., angle **1604** is about 0°). Alternatively, second outer shell **1710** may be rotated (e.g., clockwise or counterclockwise) relative to first outer shell **1706** such as to offset a major axis **1534** of an elliptical shape of a structure **1520** to be formed with a major axis **1536** of an elliptical shape of custom shell **1506** to be formed (e.g., angle **1604** is between about -90° and 90°).

[0190] In some examples, mold **1700** further comprises an insert **1718** that may be positioned within mold **1700** to extend through first interior **1708** and second interior **1712**. In the illustrated example, insert **1718** includes a first section **1720-1** configured to form an inner shell of first portion **1702** of mold **1700**. To illustrate, first section **1720-1** extends through first interior **1708** and is adapted to form sound tube **1508** of SD part **1504**. Insert **1718** further includes a second section **1720-2** configured to form an inner shell of second portion **1704** of mold **1700**. To illustrate, second section **1720-2** extends through second interior **1712** and is adapted to form opening **1532** and cavity **1528** of structure **1520**. In the illustrated example, second section **1720-2** includes a plurality of fins **1722** extending outwardly about a circumference of second section **1720-2** such as to form radial projections **1524** of mechanical element **1502**, though second section **1720-2** may have any suitable shape for forming mechanical elements **1502** having various

geometries as described herein.

[0191] In some examples, insert **1718** of mold **1700** may have a standard shape such for forming sound tube **1508**, opening **1532** and/or cavity **1528** for multiple ITE parts **1500**, **1600**. In such instances where insert **1718** has a standard shape, insert **1718** may be reused for forming multiple ITE parts **1500**, **1600**. In some examples, insert **1718** may be made from a reusable rigid material, such as metal and/or plastic (e.g., thermoplastic). In some examples, insert **1718** may be formed such as by 3D printing, e.g., in a resin-based 3D printing technology, such as digital light processing (DLP) or another stereolithography (SLA) process.

[0192] Alternatively, one or more components of insert **1718** may have a custom shape for forming sound tube **1508**, opening **1532** and/or cavity **1528** for an individual ITE part **1500**, **1600**. For example, insert **1718** may have a custom shape in which second section **1720-2** is offset and/or rotated at an angle (e.g., angle **1602** and/or angle **1604**) relative to first section **1720-1** such as to adapt the shape of one or more of sound tube **1508**, opening **1532** and/or cavity **1528** to an individual ITE part **1500**, **1600**. In such instances where insert **1718** may have a custom shape, insert **1718** may be formed to have the custom shape. Additionally or alternatively, one or both sections **1720** of insert **1718** may be formed to have a standard shape, but are connected with each other at a desired offset and/or rotation to form a mechanical element **1502** having a desired offset and/or rotation relative to a SD part **1504**.

[0193] First portion **1702**, second portion **1704**, and/or insert **1718** of mold **1700** may be connected and/or held in place by any suitable connection (e.g., fasteners, guide pins, clamps, magnets, etc.). As an illustrative example, FIG. **18** schematically shows a cross-sectional view of another implementation **1800** of mold **1700** having a spacer **1802** positioned within mold **1700** such as to maintain an orientation and/or alignment of second portion **1704** of mold **1700** relative to first portion **1702** of mold **1700** as well as an orientation and/or alignment of insert **1718** relative to first portion **1702** and/or second portion **1704** of mold **1700**. For example, spacer **1802** is positioned within at least a portion of second interior **1712** of second portion **1704** (e.g., between inner shell of second section **1720-2** and second outer shell **1710**) and at least a portion of first interior **1708** of first portion **1702** (e.g., between inner shell of first section **1720-1** and first outer shell **1706**). Spacer **1802** may thereby prevent rotation and/or an offset between first portion **1702** and second portion **1704** and/or between insert **1718** and first portion **1702** and/or second portion **1704**.

[0194] In some examples, spacer **1802** is placed over insert **1718** such that first outer shell **1706** and/or second outer shell **1710** may then be placed over spacer **1802** (e.g., first outer shell **1706** and/or second outer shell **1710** may be formed by two or more parts that are positioned over spacer **1802** and insert **1718**). In some examples, spacer **1802** is formed from a dissolvable material that may be configured to dissolve when a material for forming mechanical element **1502** and/or SD part **1504** is introduced into mold **1700**. Still other suitable configurations for mold **1700** may be used.

[0195] FIG. **19** shows an illustrative method **1900** for forming an ITE part **1500**, **1600** using a mold **1700**. While FIG. **19** illustrates exemplary operations according to one embodiment, other embodiments may omit, add to, reorder, and/or modify any of the operations shown in FIG. **19**. Moreover, each of the operations depicted in FIG. **19** may be performed in any of the ways described herein.

[0196] As shown, method **1900** includes, at operation **1902**, providing a mold **1700** comprising a first portion **1702** having a first interior **1708** shaped to form a custom shell **1506** of an SD part **1504** and a second portion **1704** having a second interior **1712** shaped to form a structure **1520** of a mechanical element **1502**. For example, second portion **1704** of mold **1700** may be positioned at a medial end **1716** of first portion **1702** such that first interior **1708** is connected with second interior **1712**. First portion **1702** and/or second portion **1704** may be formed such as by 3D printing, e.g., in a resin based 3D printing technology, such as digital light processing (DLP) or another stereolithography (SLA) process. First interior **1708** of first portion **1702** may be customized such

as based on the impressions of the individual ear canal, while second interior **1712** of second portion **1704** may have a standard shape.

[0197] In some examples, at least second portion **1704** of mold **1700** comprises an inner shell (e.g., second section **1720-2** of insert **1718**), an outer shell (e.g., second outer shell **1710**), and a spacer **1802** positioned between the inner shell and the outer shell such that spacer **1802** is configured to align the inner shell with the outer shell (e.g., to prevent rotation of the inner shell within the outer shell). In some examples, spacer **1802** may additionally extend within at least a portion of first portion **1702** of mold **1700** such as to align an inner shell (e.g., first section **1720-1** of insert **1718**) of first portion **1702** with an outer shell (e.g., first outer shell **1706**) of first portion **1702** (e.g., to prevent rotation of the inner shell with the outer shell). Spacer **1802** may further align second portion **1704** of mold **1700** with first portion **1702** of mold **1700** such as a desired offset (e.g., angle **1602**) and/or rotation (e.g., angle **1604**). Spacer **1802** may further be configured to maintain the orientation of first portion **1702** relative to second portion **1704** such as by preventing rotation of second portion **1704** relative to first portion **1702**.

[0198] Method **1900** further includes, at operation **1904**, forming mechanical element **1502** integrally with custom shell **1506** of SD part **1504** using mold **1700**. For example, mechanical element **1502** and custom shell **1506** may be cast such as by injecting first interior **1708** and second interior **1712** of mold **1700** with a material for forming mechanical element **1502** and custom shell **1506** (e.g., mechanical element **1502** and custom shell **1506** are formed with the same material, such as silicone). In instances where mold **1700** includes an insert **1718**, the material may be provided between insert **1718** and first outer shell **1706** and second outer shell **1710** such as to form sound tube **1508**, opening **1532**, and/or cavity **1528** within mechanical element **1502** and/or SD part **1504**. In instances where mold **1700** includes a spacer **1802**, spacer **1802** may be configured to dissolve when the material is introduced into mold **1700**.

[0199] After the material is introduced into mold **1700**, the material within mold **1700** may be cured (e.g., for about 10 minutes, such as in a pressure over at about 80° C.). After curing, mechanical element **1502** and custom shell **1506** may be removed from mold **1700** such that mechanical element **1502** is integrally formed with custom shell **1506**. In some examples, first portion **1702** of mold **1700** is broken away and/or dissolved from custom shell **1506**, while second portion **1704** of mold **1700** may be removed from mechanical element **1502** such that second portion **1704** may be reused. For example, second portion **1704** of mold **1700** may be reused such as by connecting second portion **1704** of mold **1700** with another first portion **1702** and forming another mechanical element **1502** integrally with another custom shell **1506** having a different shape. Still other suitable methods for forming structure **1520** of mechanical element **1502** integrally with custom shell **1506** may be used. To illustrate, ITE parts **1500**, **1600** may be formed by 3D printing a mechanical element **1502** integrally with a custom shell **1506**.

[0200] While the principles of the disclosure have been described above in connection with specific devices and methods, it is to be clearly understood that this description is made only by way of example and not as limitation on the scope of the invention. The above described embodiments are intended to illustrate the principles of the invention, but not to limit the scope of the invention. Various other embodiments and modifications to those embodiments may be made by those skilled in the art without departing from the scope of the present invention that is solely defined by the claims. In the claims, the word “comprising” does not exclude other elements or steps, and the indefinite article “a” or “an” does not exclude a plurality. A single processor or controller or other unit may fulfil the functions of several items recited in the claims. The mere fact that certain measures are recited in mutually different dependent claims does not indicate that a combination of these measures cannot be used to advantage. Any reference signs in the claims should not be construed as limiting the scope.

Claims

1. A hearing device configured to be worn at an ear of a user, the hearing device comprising: a sound delivery (SD) part configured to be at least partially inserted into an ear canal, the SD part comprising a custom shell having a shape at least partially customized to the ear canal and an electroacoustic transducer housed within the custom shell; and a mechanical element connected to a medial end of the custom shell of the SD part, the mechanical element comprising a structure enclosing a cavity with an opening facing the SD part such that the cavity is acoustically coupled to the electroacoustic transducer, wherein the structure comprises a section in which thinner portions and thicker portions are alternately arranged around an axis of the structure.
2. The hearing device of claim 1, wherein the structure of the mechanical element is standardized between multiple hearing devices.
3. The hearing device of claim 1, wherein the mechanical element is mountable to the medial end of the SD part.
4. The hearing device of claim 1, wherein the mechanical element is integrally formed with the medial end of the SD part.
5. The hearing device of claim 1, wherein the SD part comprises a sound tube having an axis in which direction sound is propagated, wherein the axis of the structure is aligned at an orientation that is within 20° relative to the axis of the sound tube when the mechanical element is connected to the SD part.
6. The hearing device of claim 5, wherein the orientation of the axis of the structure relative to the axis of the sound tube is adapted based on a geometry of the ear canal.
7. The hearing device of claim 5, wherein the structure is rotatable about the axis of the sound tube.
8. The hearing device of claim 7, wherein the rotation of the structure about the axis of the sound tube is adapted based on a geometry of the ear canal.
9. The hearing device of claim 1, wherein the thinner portions of the mechanical element are configured to transmit sound through the structure of the mechanical element.
10. The hearing device of claim 1, wherein a medial end of the mechanical element is closed.
11. The hearing device of claim 1, wherein the structure has a star-shaped cross section.
12. The hearing device of claim 1, wherein the structure has a cross section with a plurality of projections extending in a radial direction, wherein each of the projections comprises at least two thinner portions.
13. The hearing device of claim 1, wherein the thinner portions have a width between adjoining thicker portions, and the thicker portions have a width between adjoining thinner portions, wherein the width of at least one of the thinner portions exceeds the width of the adjoining thicker portions by at least a factor of two.
14. The hearing device of claim 1, further comprising a sealing member configured for sealing against an ear canal wall, wherein the sealing member is integrally formed with the element.
15. A hearing device configured to be worn at an ear of a user, the hearing device comprising: a sound delivery (SD) part configured to be at least partially inserted into an ear canal, the SD part comprising a custom shell having a shape at least partially customized to the ear canal and an electroacoustic transducer housed within the custom shell; and a mechanical element integrally formed with a medial end of the custom shell of the SD part, the mechanical element comprising a structure enclosing a cavity with an opening facing the SD part such that the cavity is acoustically coupled to the electroacoustic transducer, wherein the structure comprises a section in which thinner portions and thicker portions are alternately arranged around an axis of the structure.
16. A method of manufacturing a hearing device configured to be worn at an ear of a user, the hearing device comprising a sound delivery (SD) part configured to be at least partially inserted into an ear canal, the SD part comprising a custom shell having a shape at least partially

customized to the ear canal and an electroacoustic transducer housed within the custom shell, and a mechanical element integrally formed with a medial end of the custom shell of the SD part, the mechanical element comprising a structure enclosing a cavity with an opening facing the SD part such that the cavity is acoustically coupled to the electroacoustic transducer, wherein the structure comprises a section in which thinner portions and thicker portions are alternatingly arranged around an axis of the structure, the method comprising: providing a mold comprising a first portion having a first interior shaped to form the custom shell of the SD part and a second portion having a second interior shaped to form the structure of the mechanical element, the second portion positioned at a medial end of the first portion such that the first interior is connected with the second interior; and forming the mechanical element integrally with the custom shell of the SD part using the mold.

17. The method of claim 16, further comprising reusing the second portion of the mold by connecting the second portion of the mold with another first portion and forming another mechanical element integrally with another custom shell having a different shape.

18. The method of claim 16, wherein the second portion of the mold comprises an inner shell, an outer shell, and a spacer positioned between the inner shell and the outer shell, the spacer configured to prevent rotation of the second portion relative to the first portion.

19. The method of claim 18, wherein the spacer is dissolvable.

20. The method of claim 16, wherein the second portion is aligned with the first portion such that an axis of the second portion is aligned within 20° relative to an axis of the first portion.
